

Interview Preparation Guide

300 Expert Q&A with Detailed Answers

GCP Fundamentals • IAM • Compute Engine • GKE • Networking • Storage • Security
App Engine • Cloud Functions • Monitoring & Logging • Scenario-based Questions

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Section 1: GCP Fundamentals

30 questions with comprehensive, interview-focused answers.

Q1.

What is Google Cloud Platform (GCP)?

Answer:

GCP is Google's suite of cloud computing services running on the same global infrastructure Google uses for Search, Gmail, and YouTube. It offers 200+ services spanning compute, storage, networking, ML, and analytics, enabling organizations to build, deploy, and scale applications worldwide with enterprise-grade reliability and security.

Q2.

What are the core service categories in GCP?

Answer:

Compute: Compute Engine (VMs), GKE (Kubernetes), App Engine (PaaS), Cloud Run (serverless containers), Cloud Functions (FaaS). Storage: Cloud Storage (objects), Persistent Disk (block), Filestore (NFS). Databases: Cloud SQL, Spanner, Firestore, Bigtable, AlloyDB. Networking: VPC, Load Balancing, Cloud CDN. Analytics: BigQuery, Dataflow, Pub/Sub. AI/ML: Vertex AI, AutoML.

Q3.

What is a GCP Project?

Answer:

A GCP Project is the fundamental organizational and billing unit. It groups resources, isolates them for security, and ties them to a billing account. Each project has a unique Project ID (immutable, globally unique), Project Name (display only), and Project Number (auto-assigned). All resources belong to exactly one project and IAM policies are scoped per project.

Q4.

What is the GCP Resource Hierarchy?

Answer:

Organization □ Folders □ Projects □ Resources. Organization is the root, linked to Google Workspace or Cloud Identity domain. Folders group projects by business unit or environment (prod/dev). Projects contain resources. IAM policies set at higher levels inherit downward to all children, enabling consistent governance across the entire organization.

Q5.

What is Cloud Identity?

Answer:

Cloud Identity is Google's Identity as a Service (IDaaS) offering. It manages user accounts and groups for GCP access without requiring Google Workspace. Features: SSO, multi-factor authentication (MFA), device management, and user lifecycle management. It's the identity foundation for GCP IAM in organizations that don't use Google Workspace.

Q6.

What is the difference between a region and a zone in GCP?

Answer:

A Region is a specific geographic location (e.g., us-central1) containing 3+ isolated Zones. A Zone is a deployment area within a region (e.g., us-central1-a), physically isolated from other zones. Zones protect against single points of failure; multi-region protects against regional outages. VMs and disks are zonal; Cloud Storage can be regional or multi-regional.

Q7.

What are multi-region locations in GCP?

Answer:

Multi-region locations for services like Cloud Storage span multiple geographic regions: US (multiple US regions), EU (European regions), ASIA (Asian regions). Data is replicated across at least two geographically separate regions, providing 99.95%+ SLA and disaster recovery. Higher cost than single-region but critical for globally accessed, business-critical data.

Q8.

What is the GCP Free Tier?

Answer:

GCP Free Tier has two components: (1) Always Free — permanent free quotas including 1 e2-micro VM/month in us-east1, 5 GB Cloud Storage, 10 GB BigQuery queries/month, and limits on Cloud Functions, Pub/Sub, and Firestore. (2) Free Trial — \$300 credit valid for 90 days for new accounts to explore all services. Always Free persists beyond the trial period.

Q9.

What is Cloud Shell?

Answer:

Cloud Shell is a browser-based, authenticated command-line environment with 5 GB persistent home directory storage. Pre-installed tools: gcloud, kubectl, git, Docker, Helm, Terraform, Python, Node.js. Runs on a managed Debian VM. No local setup required — open from any browser. Ideal for quick GCP management, development, and learning without configuring a local environment.

Q10.

What is the gcloud CLI?

Answer:

gcloud is the primary CLI tool for GCP. Key capabilities: authentication (gcloud auth login), project/config management (gcloud config set project), resource management (gcloud compute instances create), and scripting CI/CD pipelines. Part of the Google Cloud SDK alongside gsutil (Cloud Storage), bq (BigQuery), and kubectl (GKE). Install SDK locally or use Cloud Shell.

Q11.

What are Labels in GCP?

Answer:

Labels are key-value metadata pairs (e.g., env:prod, team:backend, cost-center:eng) attached to GCP resources. Uses: cost allocation (filter billing exports by label), resource organization, automation triggers (e.g., lifecycle scripts targeting label-selected resources). Labels are not access controls — they're purely metadata. Up to 64 labels per resource.

Q12.

What is Cloud Deployment Manager?

Answer:

Deployment Manager is GCP's native Infrastructure as Code (IaC) service. Define resources in YAML, Python, or Jinja2 templates and Deployment Manager creates/updates resources declaratively. It manages dependency ordering and provides rollback on failures. For most new projects, Terraform is preferred for its multi-cloud support, larger community, and richer ecosystem.

Q13.

What is the difference between Capex and Opex in the cloud context?

Answer:

Capex (Capital Expenditure) is upfront investment in hardware with long depreciation cycles — the traditional on-premises model. Opex (Operational Expenditure) is consumption-based pay-as-you-go — the cloud model. GCP converts infrastructure spend from Capex to Opex, eliminating upfront costs, enabling elastic scaling, and aligning spend with actual business activity.

Q14.

What is a Committed Use Discount (CUD)?

Answer:

CUDs provide up to 57% discount in exchange for committing to use specific GCP resources for 1 or 3 years. Resource-based CUDs: commit to vCPU and memory amounts for Compute Engine and GKE. Spend-based CUDs: commit to a minimum spend for Cloud SQL and other services. Commitments are per region. Analyze historical usage in billing to identify optimal commitment levels.

Q15.

What are Sustained Use Discounts (SUD)?

Answer:

SUDs are automatic discounts applied when Compute Engine VM instances run for more than 25% of the billing month. GCP progressively increases the discount as usage grows, reaching up to 30% off on-demand pricing for full-month VMs. No commitment required — automatically applied to eligible N1, N2, N2D, E2, and custom machine types. SUDs and CUDs can stack.

Q16.

What is GCP Marketplace?

Answer:

GCP Marketplace is a catalog of pre-configured software solutions from Google and ISV partners. Offerings: VM images (LAMP, Jenkins, SAP), Kubernetes apps (Elasticsearch, MongoDB), managed services (Datadog, Terraform Enterprise), and datasets. Deploy production-ready stacks in minutes with billing integrated into your GCP bill. Simplifies procurement for complex third-party software.

Q17.

What is Anthos?

Answer:

Anthos is GCP's hybrid and multi-cloud platform built on GKE. It enables running containerized workloads on GCP, on-premises (Anthos clusters on VMware/bare metal), and on AWS/Azure with consistent management. Anthos Config Management provides GitOps-based policy enforcement. Anthos Service Mesh (based on Istio) provides observability and security across all environments.

Q18.

What is Cloud Interconnect?

Answer:

Cloud Interconnect provides high-bandwidth private connectivity from on-premises to GCP VPCs. Dedicated Interconnect: direct physical connection to Google's network (10 Gbps or 100 Gbps circuits). Partner Interconnect: connection via a supported service provider (50 Mbps to 50 Gbps). Both bypass the public internet, offering predictable latency and lower data transfer pricing than internet-based connections.

Q19.

What is the difference between Cloud Interconnect and Cloud VPN?

Answer:

Cloud VPN: IPsec tunnels over public internet, up to 3 Gbps per tunnel, lower cost, suitable for moderate bandwidth. Cloud Interconnect: private dedicated circuits, up to 200 Gbps, no internet traversal, no encryption overhead, predictable latency, better SLA. Choose VPN for cost-sensitive or lower-bandwidth requirements; Interconnect for high-throughput, latency-sensitive enterprise workloads.

Q20.

What is Customer-Managed Encryption Keys (CMEK)?

Answer:

CMEK lets customers use encryption keys they own (stored in Cloud KMS) to protect GCP resources including Cloud Storage, BigQuery, Compute Engine disks, and Cloud SQL. You control key rotation schedules and can revoke access by disabling keys, preventing Google from accessing your data. CMEK satisfies compliance requirements needing customer key control beyond default Google-managed encryption.

Q21.

What are GCP SLAs?

Answer:

GCP SLAs define monthly uptime commitments: Compute Engine multi-zone MIG: 99.99%; single-instance SSD VM: 99.9%; Cloud Storage multi-region: 99.95%; Cloud SQL HA: 99.95%; GKE: 99.95% (control plane); Cloud Spanner multi-region: 99.999%. SLA credits (10–50% of fees) are issued automatically when Google fails to meet commitments. Plan architectures around SLA tiers.

Q22.

What is a Service Account?

Answer:

A Service Account is a non-human identity (e.g., myapp@project.iam.gserviceaccount.com) used by applications, VMs, Cloud Functions, and other workloads to authenticate to GCP APIs. Grant IAM roles to service accounts to define what they can access. Best practice: one service account per workload with minimum required permissions. Avoid using user accounts for applications.

Q23.

What is the difference between Cloud Audit Logs and Cloud Logging?

Answer:

Cloud Logging is the unified log management service — ingests, stores, and provides search/analysis for all log types from GCP services, applications, and agents. Cloud Audit Logs are a specific log type within Cloud Logging that records API-level administrative activity (Admin Activity logs), data access (Data Access logs), and system events for compliance and security investigation.

Q24.

What is Google Cloud's Shared Responsibility Model?

Answer:

Google is responsible for security of the cloud: physical infrastructure, hardware, network fabric, and fully managed service layers (software patches for managed databases, GKE control plane). Customers are responsible for security in the cloud: data classification, IAM configuration, OS patching (for IaaS VMs), application security, network firewall rules, and encryption key management.

Q25.

What is Cloud Asset Inventory?

Answer:

Cloud Asset Inventory provides a complete, historical record of GCP resource metadata and IAM policies across projects, folders, and organizations. Features: bulk asset export, real-time change feed via Pub/Sub, policy analysis (who has access to what), and search across resource types. Essential for governance, compliance auditing, and understanding your GCP estate at scale.

Q26.

What is Organization Policy Service?

Answer:

Organization Policy Service enforces constraints on GCP resource configurations centrally. Example constraints: `compute.requireOsLogin` (enforce OS Login on all VMs), `iam.disableServiceAccountKeyCreation` (prevent SA key creation), `compute.restrictCloudFunctions` to specific regions. Policies set at Org/Folder level override project-level settings, providing guardrails that developers cannot bypass.

Q27.

What is Cloud Billing?

Answer:

Cloud Billing accounts are linked to GCP projects and track resource costs. A billing account can be linked to multiple projects. Key IAM roles: Billing Account Administrator, Billing Account Viewer, Project Billing Manager. Features: budgets and alerts (notify when spend exceeds threshold), commitment purchase, billing export to BigQuery for detailed cost analysis and chargebacks.

Q28.

What is the purpose of gcloud config configurations?

Answer:

gcloud configurations store named sets of properties: account, project, region, zone. Switch between configurations with: `gcloud config configurations activate CONFIG_NAME`. Useful when managing multiple projects or accounts (e.g., switch from dev-project to prod-project instantly). List configurations: `gcloud config configurations list`. Avoids needing `--project` and `--region` on every command.

Q29.

What are GCP Spot VMs?

Answer:

Spot VMs (formerly Preemptible VMs) use excess GCP capacity at up to 91% discount vs standard pricing. GCP can reclaim Spot VMs at any time with a 30-second shutdown signal. Duration: maximum 24 hours per instance. Ideal for fault-tolerant batch workloads, data processing, rendering, and CI/CD agents. Not suitable for stateful applications or services requiring continuous availability.

Q30.

What is Cloud Endpoints?

Answer:

Cloud Endpoints is an API management system for building, securing, and monitoring APIs. Uses an Extensible Service Proxy (ESP or ESPv2) to enforce: API key authentication, JWT/OAuth2 validation, rate limiting, and request logging. Supports OpenAPI and gRPC. Works with backends on App Engine, GKE, Compute Engine, or Cloud Run. Generates API portal documentation automatically.

Section 2: IAM — Identity & Access Management

30 questions with comprehensive, interview-focused answers.

Q31.

What are the IAM principal types in GCP?

Answer:

Principals (members) that can be granted IAM roles: Google Account (user@gmail.com — individual human), Service Account (app@project.iam.gserviceaccount.com — workload identity), Google Group (group@domain.com — collection of accounts), Google Workspace Domain (all users in a domain), Cloud Identity Domain, allAuthenticatedUsers (any authenticated Google account), allUsers (truly public, including anonymous).

Q32.

What is the difference between IAM Roles and IAM Policies?

Answer:

An IAM Role is a named collection of permissions (e.g., roles/storage.objectViewer bundles storage.objects.get and storage.objects.list). An IAM Policy is a binding that grants specific roles to specific principals on a specific resource. The policy document contains: {bindings: [{role: 'roles/...', members: ['user:...']}]}. Roles define what; policies define who gets what on which resource.

Q33.

What are the three types of IAM Roles?

Answer:

Basic Roles: Viewer (read-only), Editor (read+write), Owner (full control including IAM) — coarse-grained, avoid in production. Predefined Roles: created and maintained by Google, service-specific and fine-grained (e.g., roles/compute.instanceAdmin.v1). Custom Roles: user-defined combination of permissions following exact least-privilege requirements. Always prefer predefined or custom over basic roles.

Q34.

What is IAM Conditions?

Answer:

IAM Conditions add attribute-based access control to role bindings using Google's Common Expression Language (CEL). Conditions evaluate resource attributes (name, type, tags), time/date, and request origin (IP address). Example: grant Storage Object Viewer only for objects with name prefix 'public/' or only on weekdays 9am-5pm. Enables time-limited and resource-scoped access without separate IAM bindings.

Q35.

What is the principle of least privilege in GCP?

Answer:

Least privilege means granting only the minimum permissions required for a task. Implementation: use predefined roles over basic roles, create Custom Roles for fine-grained control, assign roles at the lowest necessary resource level (bucket vs project), use IAM Recommender to identify and remove excess permissions, audit bindings regularly, and set time-bound access with IAM Conditions for temporary needs.

Q36.

What is Workload Identity Federation?

Answer:

Workload Identity Federation allows external workloads (GitHub Actions CI/CD, AWS Lambda, Azure functions, on-premises apps) to authenticate to GCP without long-lived service account keys. External identities are exchanged for short-lived GCP credentials via attribute conditions and impersonation. Eliminates key rotation burden and reduces key leakage risk. Supported identity providers: GitHub, AWS, Azure, OIDC-compliant providers.

Q37.

What is the IAM Recommender?

Answer:

IAM Recommender uses ML analysis of 90-day actual API usage patterns to suggest removing excess permissions. It identifies: roles granted but never exercised, and suggests smaller replacement roles. Recommendations show the specific actions affected and security improvement score. Integrates with SCC. Apply recommendations to enforce least privilege at scale without manual permission analysis.

Q38.

How do you grant a service account access to a specific Cloud Storage bucket?

Answer:

Resource-level binding: `gcloud storage buckets add-iam-policy-binding gs://BUCKET_NAME --member=serviceAccount:SA_EMAIL --role=roles/storage.objectAdmin`. This is preferred over project-level grants as it follows least privilege. Verify: `gcloud storage buckets get-iam-policy gs://BUCKET_NAME`. Always grant at the most specific resource level that satisfies the use case.

Q39.

What is the difference between `allUsers` and `allAuthenticatedUsers`?

Answer:

`allUsers`: grants access to everyone on the internet, including unauthenticated requests — truly public. `allAuthenticatedUsers`: grants access to any user signed into any Google account (Gmail, Workspace, Cloud Identity) but not anonymous users. Use `allUsers` for public websites or open data. Never use either for sensitive or internal data. Organization Policy can restrict these from being granted.

Q40.

What are Service Account Keys and why are they risky?

Answer:

Service Account Keys are downloadable JSON credential files that authenticate as the service account from anywhere with no IP restriction and no expiry by default (max 10 years). Risks: accidentally committed to Git, exposed in CI/CD logs, stolen from developer machines, difficult to detect misuse. Best practice: avoid keys entirely. Use Workload Identity Federation, Workload Identity (GKE), or ADC instead.

Q41.

What is Application Default Credentials (ADC)?

Answer:

ADC is a credential discovery strategy used by GCP client libraries. Search order: (1) `GOOGLE_APPLICATION_CREDENTIALS` env variable pointing to a key file; (2) `gcloud application-default credentials` (developer workstations); (3) Attached service account via GCP metadata server (VMs, Cloud Run, Cloud Functions, GKE). ADC enables writing code that works in all environments without changing credential handling.

Q42.

What is Cloud Identity-Aware Proxy (IAP)?

Answer:

IAP implements application-level access control for web applications on App Engine, Compute Engine, and GKE. It sits in front of the application and authenticates users via Google Identity, then checks IAM policies before proxying requests. Only principals with roles/`iap.httpsResourceAccessor` reach the app. Replaces VPN for securing internal tools — employees access from anywhere without VPN.

Q43.

What is an IAM Deny Policy?

Answer:

Deny Policies explicitly block specified permissions for specified principals, overriding any allow grants. Evaluated before allow policies in the IAM evaluation flow. Use cases: ensure no one (including Owners) can delete a production database; prevent all users from disabling audit logs; create non-bypassable guardrails. Useful for compliance requirements that must hold even for highly privileged users.

Q44.

How does IAM inheritance work in the resource hierarchy?

Answer:

IAM policies are additive downward: policies set at Organization level apply to all Folders, Projects, and Resources below. Project-level policies add to (never restrict) folder/org policies. Resource-level policies add to project policies. A principal with Owner at the Org level has Owner on every project and resource. This is why granting broad roles at high levels is risky — scope them as low as possible.

Q45.

What is the difference between a Service Account and a User Account?

Answer:

User Accounts represent humans — authenticated with passwords and MFA, managed through Google Workspace or Cloud Identity, cannot be impersonated programmatically. Service Accounts represent non-human workloads — no password, authenticated with keys or automatically via GCP metadata server. Service accounts are IAM resources (you manage them with `gcloud iam service-accounts` commands), not directory objects.

Q46.

What is Domain Restricted Sharing?

Answer:

The `iam.allowedPolicyMemberDomains` Organization Policy constraint restricts IAM bindings to only allow principals from specified Google Workspace or Cloud Identity domains. Prevents accidental external sharing by blocking granting access to gmail.com or other external accounts. Essential for organizations requiring strict data access controls – enforce at organization level to protect all projects.

Q47.

What is a Google Group and why use it for IAM?

Answer:

A Google Group is a named collection of user and service accounts with a single email address. Granting IAM roles to groups instead of individuals is a best practice: add/remove a user from the group to instantly change their GCP access across all projects. Easier to audit (who's in the group?), scales to hundreds of users without managing hundreds of individual bindings.

Q48.

What is the difference between Cloud Identity and Google Workspace for GCP?

Answer:

Google Workspace includes collaboration tools (Gmail, Drive, Meet, Calendar) plus identity management at a higher per-user cost. Cloud Identity is identity-only (user accounts, groups, device management, SSO) with a free tier for basic features. Both enable managed accounts for GCP IAM. Organizations not needing Google Workspace's collaboration tools should use Cloud Identity for cost-effective GCP identity management.

Q49.

How do you rotate a service account key safely?

Answer:

Step 1: Create a new key: `gcloud iam service-accounts keys create new-key.json --iam-account=SA_EMAIL`. Step 2: Update your application/CI system to use the new key. Step 3: Deploy and verify the application works with the new key. Step 4: Delete the old key: `gcloud iam service-accounts keys delete OLD_KEY_ID --iam-account=SA_EMAIL`. Automate rotation using Secret Manager's rotation feature with 90-day maximum key age.

Q50.

What is VPC Service Controls?

Answer:

VPC Service Controls creates security perimeters around GCP API services (BigQuery, Cloud Storage, etc.) to prevent data exfiltration. Even users with IAM permissions cannot access perimeter-protected resources from outside the perimeter. Access policies define which projects are inside the perimeter. Access levels define conditions for bridging the perimeter (device policy, IP range, identity). Protects against compromised credentials used from outside.

Q51.

What is the roles/iam.serviceAccountTokenCreator role?

Answer:

This role allows the grantee to generate short-lived OAuth2 access tokens and ID tokens for a service account, effectively impersonating it. Used for: service account chaining (A acts as B), delegated signing in serverless architectures. Grant very carefully — holder gains all permissions of the target SA. Audit all principals with this role regularly. Prefer direct IAM grants over SA impersonation when possible.

Q52.

What is Attribute-Based Access Control (ABAC) in GCP?

Answer:

ABAC in GCP is implemented through IAM Conditions (CEL expressions) and Resource Manager Tags. Instead of granting access based purely on identity + role, ABAC makes access decisions based on resource attributes (tags, name prefix), time, and request context. Example: allow VM access only to instances tagged env:dev. Enables dynamic, context-sensitive access control without maintaining separate roles per resource group.

Q53.

How do you audit IAM changes in GCP?

Answer:

Cloud Audit Logs — Admin Activity logs — automatically record all IAM policy changes (SetIamPolicy calls) with who made the change, when, and what changed. Retained 400 days, cannot be disabled, zero cost. Query in Logs Explorer: resource.type='project' AND protoPayload.methodName:'SetIamPolicy'. Export to BigQuery for trend analysis. Set up log-based metric alerts for unexpected IAM changes.

Q54.

What is a Custom Role and when should you create one?

Answer:

Create Custom Roles when: predefined roles grant more permissions than needed (violating least privilege), or when you need a precise combination of permissions not available in any single predefined role. Define via YAML with a list of permissions. Create at project or organization level. Note: Custom Roles require maintenance — review when Google launches new API versions or deprecates permissions. Don't duplicate predefined roles unnecessarily.

Q55.

What is the difference between Owner, Editor, and Viewer basic roles?

Answer:

Viewer (roles/viewer): read-only access to all GCP resources in the project. Editor (roles/editor): all Viewer permissions plus create, update, delete most resources. Owner (roles/owner): all Editor permissions plus manage IAM policies, manage billing, and delete the project. All three grant across-the-board access to ALL services — avoid in production. Use specific predefined or custom roles instead.

Q56.

What is Access Context Manager?

Answer:

Access Context Manager defines Access Levels — conditions for granting privileged access based on: IP address/range, device policy (corporate-managed, certificate, OS), user identity, and geographic location. Access Levels are used by: VPC Service Controls (bridge perimeter for compliant requests), IAP (enforce device health for app access), and IAM Conditions. Implements BeyondCorp Zero Trust access model components.

Q57.

What is Security Command Center (SCC)?

Answer:

SCC is GCP's centralized security and risk management platform. Standard tier: basic asset inventory and vulnerability scanning. Premium tier adds: Event Threat Detection (detects active attacks from logs), Container Threat Detection, VM Threat Detection, Security Health Analytics (misconfigurations), Web Security Scanner, Compliance dashboards (CIS, PCI, HIPAA, NIST). Aggregates findings from Google and partner security tools for unified view.

Q58.

How do you prevent a service account from being misused?

Answer:

Preventive controls: create dedicated per-workload service accounts (not shared default SA), grant minimum required roles only, disable unused service accounts (gcloud iam service-accounts disable SA_EMAIL), set iam.disableServiceAccountKeyCreation org policy, set iam.automaticIamGrantsForDefaultServiceAccounts constraint to prevent auto Editor grants, audit SA usage in Cloud Audit Logs. Delete service accounts when workloads are decommissioned.

Q59.

What is the purpose of Resource Manager Tags vs Labels?

Answer:

Labels: key-value metadata for organization, filtering, and billing allocation — no access control impact. Resource Manager Tags: key-value pairs stored in Resource Manager (not as metadata on resources) that can be referenced in IAM Conditions and Organization Policy constraints to control access and apply policy. Tags require the Tag Administrator IAM role to manage and enable policy-driven governance based on resource classification.

Q60.

What are Privileged Access Management best practices in GCP?

Answer:

Best practices: use Google Groups for IAM (not individuals), avoid Basic roles in production, implement just-in-time access using time-bound IAM Conditions, require MFA for all admin accounts (enforce via Cloud Identity), enable Org-level Admin Activity audit logs, use IAM Recommender quarterly, separate production and non-production projects with different IAM bindings, use Deny Policies for critical guardrails, and rotate service account keys every 90 days.

Section 3: Compute Engine

30 questions with comprehensive, interview-focused answers.

Q61.

What is Google Compute Engine?

Answer:

Compute Engine is GCP's Infrastructure as a Service (IaaS), providing virtual machines on Google's global infrastructure. Features: 100+ predefined machine types, custom machine types, live migration during host maintenance, automatic restart policies, persistent and local SSD storage, global load balancing, managed instance groups with auto-scaling, and per-second billing (1-minute minimum).

Q62.

What are Compute Engine machine families?

Answer:

General Purpose: E2 (cost-optimized, shared-core options), N2/N2D (balanced performance, Intel/AMD), N1 (legacy flexible). Compute Optimized: C2 (3.8 GHz Intel, highest single-thread), C2D (AMD EPYC). Memory Optimized: M1/M2 (up to 12 TB RAM for SAP HANA). Accelerator Optimized: A2 (NVIDIA A100 GPUs). Storage Optimized: Z3 (high IOPS local SSD). Choose based on workload characteristics — not all families support all features.

Q63.

What is a custom machine type?

Answer:

Custom machine types let you specify exact vCPU and memory rather than choosing from predefined types. Choose vCPUs (1 or even number 2–224 depending on family) and memory in 256 MB increments (0.9–8 GB per vCPU standard range). Extended Memory allows up to 6.5 GB/vCPU. Useful when predefined types are either too large (wasteful) or too small (insufficient). Custom types are slightly more expensive per resource than nearest predefined type.

Q64.

What is the difference between persistent disks and local SSDs?

Answer:

Persistent Disks: network-attached, survive VM stop/restart/deletion (independent lifecycle), replicated for durability, resizable, types from HDD to Extreme SSD. Local SSDs: physically attached to VM host, 375 GB per disk (up to 24 disks), highest performance (2.4M IOPS, 1.2 TB/s bandwidth), ephemeral – data is permanently lost when VM stops or is live-migrated. Use local SSDs for cache, temp storage, or applications managing their own replication.

Q65.

What is a Managed Instance Group (MIG)?

Answer:

A MIG is a group of identical VM instances managed as a unit using an Instance Template. Features: auto-scaling (CPU, LB utilization, custom metrics), auto-healing (replace unhealthy instances via health checks), rolling updates (update VMs without downtime), multi-zone distribution (regional MIGs span zones for HA). Zonal MIGs use one zone; Regional MIGs span multiple zones in a region.

Q66.

What is an Instance Template?

Answer:

An immutable resource defining VM configuration for MIG-created instances: machine type, boot disk image, network, service account, startup script, labels, and metadata. Once created, templates cannot be edited – create a new version to change anything. Templates are regional (but can reference global images). Used by MIGs to ensure all instances are identical. Reference by name in MIG configuration.

Q67.

How does Compute Engine auto-scaling work?

Answer:

Configure an autoscaler on a MIG specifying: min/max instance counts, cooldown period (time after scaling for metrics to stabilize), and scaling policy. Policies: CPU utilization target (scale to maintain e.g. 60% CPU), HTTP load balancing utilization (requests per instance), Pub/Sub queue depth, or custom Cloud Monitoring metrics. Autoscaler checks every 60 seconds and adds/removes instances in batches.

Q68.

What are startup and shutdown scripts in Compute Engine?

Answer:

Startup scripts execute as root at VM boot. Set via metadata key startup-script (inline script) or startup-script-url (script in Cloud Storage). Used for: installing software, configuring services, mounting extra disks, registering with orchestration systems. Shutdown scripts (metadata key shutdown-script) run when VM is stopped/deleted – drain connections, save state, deregister. Max runtime: 90 seconds for preemptible VMs.

Q69.

What is live migration in Compute Engine?

Answer:

Live Migration transparently moves running VM instances to a different physical host during planned infrastructure maintenance (hardware/software updates) without any VM downtime. The VM's state, memory, and attached disks are migrated with only brief performance impact. Most standard VMs support live migration. VMs using GPUs, TPUs, or certain local SSDs use terminate-and-restart maintenance instead.

Q70.

What are the persistent disk types in Compute Engine?

Answer:

Standard: HDD, \$0.04/GB/month, 0.75 read / 1.5 write IOPS/GB, best for sequential reads/cold data. Balanced: SSD, \$0.10/GB, 6 IOPS/GB, good balance for general workloads. SSD: \$0.17/GB, 30 IOPS/GB, production databases. Extreme: \$0.125/GB + provisioned IOPS (\$0.007/IOPS), up to 120,000 IOPS, highest-performance databases. All types replicate data for durability and can be resized online.

Q71.

What is a sole-tenant node?

Answer:

Sole-tenant nodes are physical Compute Engine servers dedicated exclusively to a single customer — no VM from other customers runs on the same hardware. Use cases: (1) Compliance requirements (HIPAA, PCI DSS) mandating physical isolation; (2) Windows Server per-core licensing (license applies to entire physical server); (3) Performance isolation for latency-sensitive applications. Configure via Node Templates and Node Groups.

Q72.

How do you create a VM snapshot?

Answer:

`gcloud compute disks snapshot DISK_NAME --snapshot-names=SNAPSHOT_NAME --zone=ZONE`. First snapshot is a full copy; subsequent snapshots are incremental (only changes). Snapshots are stored in Cloud Storage and billed per GB compressed. Use cases: point-in-time backup, disk cloning, migrating VMs to new regions, creating custom images. Schedule snapshots automatically via Snapshot Schedules (Resource Policies).

Q73.

What is a custom image in Compute Engine?

Answer:

A custom image is a boot disk image capturing OS, installed software, configuration, and data. Create from: existing VM disk, snapshot, or image file. Stored in your project's image registry. Uses: (1) Golden image for consistent VM provisioning; (2) Pre-baked dependencies to reduce startup time vs startup scripts; (3) Share across projects (--project flag). Deprecate old images and maintain a versioned image pipeline with Cloud Build.

Q74.

What is the difference between stopping and deleting a VM?

Answer:

Stopping: VM is powered off, configuration and disks are preserved. You're charged for persistent disks and reserved static IPs, not for vCPU/memory. VM can be restarted. Deleting: VM instance is permanently removed. By default, boot disk is also deleted (configure delete-on-termination=false to keep it). Ephemeral IPs are released; static external IPs remain and continue to be billed. Cannot be undone.

Q75.

What is Shielded VMs?

Answer:

Shielded VMs enhance security with three features: (1) Secure Boot: only signed, trusted code runs during boot — prevents bootkit/rootkit malware; (2) Virtual TPM (vTPM): hardware Trusted Platform Module for attestation and key storage; (3) Integrity Monitoring: measures boot sequence at each start and compares against a known-good baseline in Cloud Monitoring. Enable for sensitive workloads and compliance requirements like FedRAMP.

Q76.

What is Confidential Computing?

Answer:

Confidential VMs encrypt data in use — while being processed in RAM — using AMD EPYC CPUs with Secure Encrypted Virtualization (SEV). Memory is encrypted with a per-VM key generated and managed by the AMD hardware, inaccessible to the hypervisor or Google staff. Available on N2D and C2D machine types. Use for regulated workloads (financial data, health records, proprietary IP) requiring maximum data-in-use protection.

Q77.

What is VM Manager?

Answer:

VM Manager is an OS management suite for Compute Engine VMs running OS Config Agent. Components: (1) OS Patch Management — schedule and automate OS patching across VM fleets with compliance reporting; (2) OS Configuration Management — apply configurations via Cloud Logging; (3) OS Inventory Management — collect installed packages, OS versions, kernel details. Integrates with SCC for vulnerability reporting.

Q78.

What are Compute Engine quotas and how do you manage them?

Answer:

Quotas limit resource consumption per project/region to prevent unexpected bills and ensure fair capacity distribution. Key quotas: CPUs per region, Preemptible CPUs, In-use external IP addresses, Persistent disk SSD GB. View quotas: Console □ IAM & Admin □ Quotas. Request increases via quota request form (takes 2–3 business days typically). Monitor quota utilization in Cloud Monitoring to proactively request increases.

Q79.

What is the difference between regional and zonal resources in Compute Engine?

Answer:

Zonal: exist in a single zone (VMs, persistent disks, zonal MIGs). If the zone experiences an outage, these resources are unavailable. Regional: replicated across multiple zones in a region (regional persistent disks — synchronously replicated across 2 zones, regional MIGs — VMs spread across zones). For high availability: use regional persistent disks for stateful workloads and regional MIGs for stateless services.

Q80.

How does auto-healing work in MIGs?

Answer:

Configure auto-healing on a MIG with a Health Check defining probe (HTTP endpoint, path, expected response code) and thresholds (healthy/unhealthy consecutive check count). Initial delay (recommend 300+ seconds) gives VMs time to initialize before checks begin. The MIG polls all instances. If an instance fails health checks beyond thresholds, the MIG deletes and recreates it using the Instance Template. Respects max-unavailable limit.

Q81.

What is VM metadata and how is it used?

Answer:

VM metadata is key-value data accessible from within a VM at <http://metadata.google.internal/computeMetadata/v1/> (requires Metadata-Flavor: Google header). Auto-populated metadata: project ID, zone, service account tokens, instance ID. Custom metadata: any key-value pair set at instance or project level. Common uses: startup scripts (startup-script key), feature flags, application configuration, SSH keys. Metadata is not encrypted — don't store secrets (use Secret Manager instead).

Q82.

What is a static vs ephemeral external IP in Compute Engine?

Answer:

Ephemeral external IPs are dynamically assigned when a VM starts and released when it stops — changes on every restart. Static external IPs are reserved and persist across VM stop/restart. Reserve: `gcloud compute addresses create ADDRESS_NAME --region=REGION`. Assign to VM at creation or update. Unattached static IPs are billed. Use static IPs for: DNS records, IP whitelisting with partners, load balancer frontends.

Q83.

How do you SSH into a Compute Engine VM?

Answer:

Recommended: `gcloud compute ssh INSTANCE_NAME --zone=ZONE`. `gcloud` manages SSH key generation and upload automatically. Add `--tunnel-through-iap` to connect via IAP without external IP or firewall rules exposing port 22. Add `--project=PROJECT_ID` for cross-project access. For OS Login environments, ensure you have `roles/compute.osLogin` (non-sudo) or `roles/compute.osAdminLogin` (sudo). Works from Cloud Shell or local terminal with `gcloud` SDK.

Q84.

What is the Compute Engine default service account?

Answer:

Each project gets a default Compute Engine service account (`PROJECT_NUMBER-compute@developer.gserviceaccount.com`) that automatically receives the Editor basic role. VMs created without specifying a service account use this default. This is a major security risk — Editor grants broad project-wide access. Best practice: create dedicated service accounts with minimal roles per workload, disable the default SA if unused, set `iam.automaticIamGrantsForDefaultServiceAccounts` org policy.

Q85.

What are Access Scopes in Compute Engine?

Answer:

Access Scopes are a legacy mechanism controlling API access for VMs using the Compute Engine default service account. They pre-date IAM and limit API access via OAuth scopes (e.g., <https://www.googleapis.com/auth/cloud-platform> for full access). Best practice: set scope to cloud-platform and use IAM roles to control actual permissions. Scopes only apply to the default Compute SA — custom service accounts use IAM only.

Q86.

What is instance scheduling in Compute Engine?

Answer:

Resource Policies allow defining recurring schedules for automatically starting and stopping VM instances. Configure schedule (e.g., start Mon-Fri 8:00 AM, stop 7:00 PM) in the Compute Engine console or via gcloud. Attach to individual VMs or groups. Ideal for dev/test environments — save 60–70% cost vs always-on. Works with Compute Engine only (not Cloud Run or GKE which scale to zero).

Q87.

What is a Compute Engine Health Check?

Answer:

Health Checks probe VM instances to determine availability. Protocols: HTTP (check response code/content), HTTPS, HTTP/2, TCP, SSL, gRPC. Key parameters: check interval (default 5s), timeout (5s), healthy threshold (consecutive successes required), unhealthy threshold (consecutive failures before marking unhealthy). Used by: HTTP(S) Load Balancer (route only to healthy backends), MIG auto-healing (replace unhealthy instances).

Q88.

What is the difference between N1, N2, and E2 machine types?

Answer:

N1: First generation general-purpose, Intel Skylake/Broadwell, up to 96 vCPUs, supports GPUs and TPUs, older performance baseline, still widely used. N2: Second generation, Intel Cascade Lake, up to 128 vCPUs, 20% better price-performance than N1, supports 3-year CUDs. E2: Cost-optimized, uses both Intel and AMD via dynamic scheduling, up to 32 vCPUs, up to 50% cheaper than N1 for low-utilization workloads, no GPU support, shared-core options available.

Q89.

What is VM rightsizing?

Answer:

VM rightsizing recommendations appear in Compute Engine console □ VM instances □ Rightsizing tab. Cloud Monitoring analyzes CPU/memory utilization over the past 8 days and suggests downsizing overprovisioned VMs. Recommendations show projected cost savings. Implement by changing machine type (stop VM, change type, restart). Combine with CUDs for maximum savings. Schedule VM Manager to regularly review and act on new recommendations.

Q90.

How do you resize a Compute Engine boot disk?

Answer:

Step 1: `gcloud compute disks resize DISK_NAME --size=NEW_SIZE_GB --zone=ZONE` (can only increase size, never decrease). Step 2: For Linux – SSH into VM and run: `sudo resize2fs /dev/sda1` (ext4) or `sudo xfs_growfs /` (XFS). For Windows – use Disk Management GUI or `diskpart` to extend the volume. The GCP API resize is instant; OS-level resize takes effect immediately without reboot. Always snapshot before resizing.

Section 4: App Engine & Cloud Functions

30 questions with comprehensive, interview-focused answers.

Q91.

What is Google App Engine?

Answer:

App Engine is GCP's fully managed Platform as a Service (PaaS). Developers deploy application code and App Engine handles: server provisioning, load balancing, auto-scaling, OS patching, and runtime updates. Supports Standard and Flexible environments. Ideal for web applications and APIs needing automatic scaling without infrastructure management. First GCP product (launched 2008) with deep integration with GCP services.

Q92.

What is the difference between App Engine Standard and Flexible environments?

Answer:

Standard: language-specific sandboxed runtimes (Python, Java, Go, PHP, Node.js, Ruby), scales to zero instantly (no idle cost), free tier, cold starts in milliseconds, limited system access. Flexible: Docker containers on Compute Engine VMs, any language/runtime, no scaling to zero (minimum 1 running instance), slower scaling (minutes), full VM access, SSH debugging support. Choose Standard for cost-sensitive web apps; Flexible for custom runtimes or system-level needs.

Q93.

What is an App Engine Service?

Answer:

App Engine applications can contain multiple Services (formerly Modules), each handling distinct functionality with independent scaling and configuration. Example: default service (web frontend), api service (REST backend), worker service (background processing). Each service has its own app.yaml, runtime, scaling settings, and version history. Deploy services independently without affecting others. Traffic can be split across versions per service.

Q94.

What is app.yaml in App Engine?

Answer:

The app.yaml file configures App Engine deployments. Key fields: runtime (e.g., python310), service (default if omitted), instance_class (F1-F4 for Standard, custom for Flexible), scaling (automatic/manual/basic with limits), handlers (URL-to-script routing), env_variables (non-secret configuration), entrypoint (override default start command), and includes (import other YAML files). Every App Engine service requires this file.

Q95.

What are the scaling options in App Engine?

Answer:

Automatic Scaling: scales instances based on request volume, latency, and CPU — configure `target_cpu_utilization`, `min/max_instances`, `max_concurrent_requests`. Best for variable traffic. Basic Scaling: creates instances on demand, shuts down after idle timeout — good for sporadic workloads, lower cost but slower response time. Manual Scaling: fixed instance count always running — predictable performance, no latency from cold starts, highest cost.

Q96.

What is App Engine traffic splitting?

Answer:

Traffic splitting gradually routes incoming requests across multiple versions of a service. Split methods: IP address (consistent for same client, no cookie required), Cookie (uses GOOGAPPID cookie for precise session stickiness), Random (true percentage split, no stickiness). Canary deployments: send 1% to new version, monitor errors, increase gradually. Blue-green: send 100% to new version after testing, keep old version for instant rollback.

Q97.

What is Cloud Tasks?

Answer:

Cloud Tasks is a managed service for asynchronous task execution using HTTP task queues. Tasks are enqueued and delivered with configurable: rate limiting (max dispatches/second), retry policies (max attempts, backoff), scheduling (execute at future time), and deduplication. Targets: any authenticated HTTPS endpoint — App Engine, Cloud Run, Compute Engine, or external URLs. Evolved from App Engine Task Queues, now a standalone GCP service.

Q98.

What replaced App Engine Datastore?

Answer:

App Engine's Datastore is now Cloud Firestore in Datastore mode — fully backward-compatible API with existing Datastore code. New applications should use Cloud Firestore in Native mode for: real-time data synchronization, mobile/web SDKs, offline support, stronger querying. Datastore mode suits: existing Datastore apps, multi-region high-throughput without real-time sync. You cannot use both modes in the same project.

Q99.

What are App Engine instance classes in Standard environment?

Answer:

F-class (used with Automatic Scaling): F1 (384 MB RAM, 600 MHz), F2 (512 MB, 1.2 GHz), F4 (1 GB, 2.4 GHz), F4_1G (2 GB, 2.4 GHz). B-class (used with Manual/Basic Scaling): B1 (256 MB, 600 MHz) through B8 (4 GB, 4.8 GHz). Higher class = more resources and cost. F-class for request-driven autoscaling web apps; B-class for backend processing requiring predictable capacity.

Q100.

What is App Engine cron?

Answer:

App Engine cron jobs schedule HTTP GET requests to application URLs at defined intervals. Configure in `cron.yaml`: `schedule` ('every 5 minutes', 'every monday 09:00'), `target URL`, `description`, `retry parameters`. Cron requests include `X-Appengine-Cron: true` header – validate this in your handler. No additional cost beyond instance time. Migrate complex scheduling to Cloud Scheduler for more features (Cloud Run, Pub/Sub targets, timezone support).

Q101.

What is Cloud Functions?

Answer:

Cloud Functions is GCP's serverless, event-driven FaaS (Functions as a Service). Write single-purpose functions triggered by events. Supports: HTTP triggers (synchronous), event triggers (Pub/Sub, Cloud Storage, Firestore, Eventarc). Runtimes: Node.js, Python, Go, Java, Ruby, PHP, .NET. Auto-scales instantly; billed per invocation and compute time (GHz-seconds + GB-seconds). No server management; deploy code or a container image.

Q102.

What is the difference between Cloud Functions 1st gen and 2nd gen?

Answer:

1st gen: simpler deployment, max 9-minute timeout, max 8 GB RAM, one request per instance, legacy event system. 2nd gen (built on Cloud Run): max 60-minute timeout (HTTP), max 32 GB RAM and 8 vCPUs, concurrency up to 1000 requests per instance, Eventarc for 90+ event sources, traffic splitting, minimum instances, all Cloud Run features. New functions should default to 2nd gen for better performance, lower cold start rates, and greater flexibility.

Q103.

What triggers are available for Cloud Functions?

Answer:

HTTP Trigger: public HTTPS URL, any HTTP method, synchronous response. Eventarc Triggers (2nd gen): Cloud Storage (object created/deleted/archived/metadata updated), Pub/Sub (message published to topic), Cloud Firestore (document events), Firebase (Realtime Database, Auth, Analytics), and 90+ GCP service events via Eventarc. Legacy triggers (1st gen only): background functions for GCS and Pub/Sub without Eventarc.

Q104.

What is the execution model of Cloud Functions?

Answer:

Functions are stateless — each invocation is independent. However, GCP reuses warm instances when possible: global variables (initialized outside the function handler) persist across invocations on the same instance. Use this for: DB connection pools, loaded ML models, HTTP client objects. Avoid blocking I/O in global scope. Use async patterns (Promises, `async/await` in Node.js; `asyncio` in Python) for I/O-bound functions.

Q105.

How do you secure HTTP Cloud Functions?

Answer:

Require authentication: deploy with `--no-allow-unauthenticated`. Grant invoker access: `gcloud functions add-invoker-policy-binding FUNCTION --member=serviceAccount:SA_EMAIL --role=roles/cloudfunctions.invoker`. Callers must include a valid Google-signed ID token (Bearer token) in requests. For browser access: use IAP. Validate all input. Store secrets in Secret Manager. Use VPC connector for private resource access. Enable HTTPS only.

Q106.

What is the Cloud Functions concurrency model?

Answer:

1st gen: strict one-concurrent-request-per-instance model. Each concurrent request requires a new instance (cold or warm). For 100 concurrent requests, 100 instances run. 2nd gen: configurable concurrency up to 1000 requests per instance. For I/O-bound functions (API calls, DB queries), high concurrency dramatically reduces instance count and cold starts. For CPU-bound functions, keep concurrency lower to avoid resource contention.

Q107.

What is a cold start in Cloud Functions?

Answer:

A cold start occurs when GCP creates a new function instance because no warm instances are available. It includes: downloading function code/container, starting the runtime, running global initialization code. Duration: 100ms to 5+ seconds depending on language, dependencies, and package size. Mitigation: set `--min-instances` (keep warm instances ready), reduce package size (only import needed modules), use lighter runtimes (Go fastest, Java slowest cold start).

Q108.

What is the difference between Cloud Functions and Cloud Run?

Answer:

Cloud Functions: event-driven single functions, simpler deployment (just code), managed runtime, auto-scales to zero. Cloud Run: containerized services with full Dockerfile control, handles any workload type (not just event-driven), supports WebSockets and gRPC, longer timeouts, concurrency control, custom domains, traffic splitting. Cloud Functions 2nd gen is built on Cloud Run. Use Cloud Run when you need container flexibility; Cloud Functions for simple event-driven tasks.

Q109.

How do you pass configuration to Cloud Functions?

Answer:

Environment variables (`--set-env-vars KEY=VALUE`): visible in GCP Console, not encrypted — suitable for non-sensitive config only. Secret Manager integration (`--set-secrets KEY=SECRET:VERSION`): secrets are fetched at runtime and injected as env vars or volume-mounted files — recommended for passwords, API keys. Runtime config stored in Firestore/GCS: flexible but requires explicit fetch code. Never hardcode secrets in function source code.

Q110.

What is Cloud Functions retry behavior?

Answer:

Event-triggered functions: enable retry (`--retry` flag) for automatic retry on failure. GCP retries for up to 7 days with exponential backoff. Make functions idempotent (safe to run multiple times) — use unique event IDs to deduplicate. HTTP functions: no automatic retry by GCP (client is responsible for retries). Use dead-letter topics in Pub/Sub to capture messages that exhaust all retry attempts for manual investigation.

Q111.

What is the maximum timeout for Cloud Functions?

Answer:

1st gen: 540 seconds (9 minutes) maximum. 2nd gen HTTP-triggered: 3600 seconds (60 minutes). 2nd gen event-triggered: 600 seconds (10 minutes). Set at deployment: `--timeout=SECONDS`. Functions exceeding timeout are terminated mid-execution. For longer tasks: use Cloud Tasks (queue individual tasks), Cloud Run Jobs (batch), or Workflows (orchestrate multi-step processes with state management).

Q112.

What is VPC Connector for Cloud Functions?

Answer:

Serverless VPC Access Connector enables Cloud Functions to send traffic to private VPC resources (Cloud SQL private IP, Memorystore, internal load balancers) without internet exposure. Create connector in the same region as the function, then attach:

--vpc-connector=projects/P/locations/REGION/connectors/NAME. Route options: --vpc-connector-egress-settings=all-traffic (all outbound via VPC) or private-ranges-only (only RFC1918 ranges through VPC).

Q113.

How do you deploy a Cloud Function with gcloud?

Answer:

```
gcloud functions deploy FUNCTION_NAME --gen2 --runtime=python311
--trigger-http --entry-point=handle_request --region=asia-south1
--memory=512MB --timeout=60s --min-instances=1 --max-instances=10
--set-env-vars=ENV=prod --set-secrets=DB_PASS=db-password:latest
--no-allow-unauthenticated
--service-account=SA@PROJECT.iam.gserviceaccount.com --source=./src. For
Pub/Sub trigger: --trigger-topic=TOPIC_NAME.
```

Q114.

What is App Engine Memcache?

Answer:

App Engine Memcache provides shared in-memory caching. Shared Memcache: free, best-effort (no SLA, no dedicated capacity). Dedicated Memcache: provisioned GB capacity, SLA. Both support: get, set, delete, increment/decrement, compare-and-swap. For new applications: use Memorystore (managed Redis or Memcached) which works with all GCP compute services (not just App Engine), provides dedicated capacity, and offers Redis data structures.

Q115.

What is the App Engine Admin API?

Answer:

The App Engine Admin API (appengine.googleapis.com) enables programmatic management: deploy new versions, route traffic, configure cron jobs, and manage App Engine settings. Used internally by gcloud app deploy. CI/CD pipelines use it via service account with roles/appengine.deployer + roles/storage.objectCreator (for deployment bucket). Use gcloud app versions list/describe to inspect deployed versions.

Q116.

What are Cloud Functions pricing factors?

Answer:

Pricing per month (with always-free tier): Invocations: 2M free, then \$0.40/million. Compute Time: 400,000 GB-seconds free, then \$0.0000025/GB-second. CPU: 200,000 GHz-seconds free, then \$0.0000100/GHz-second. Networking: standard GCP egress charges apply. 2nd gen with min-instances charges for idle instances. Optimize: minimize function memory (pay for what you allocate), reduce execution duration, avoid unnecessary dependencies.

Q117.

What is Eventarc?

Answer:

Eventarc is GCP's managed event routing service delivering events in CloudEvents format (CNCF standard) from 90+ Google service sources, custom HTTP endpoints, and third-party providers to targets: Cloud Functions 2nd gen, Cloud Run, GKE, and Workflows. Replaces legacy Cloud Functions event triggers. Better observability, filtering, transformation, and replay capabilities. Use for building loosely coupled, event-driven architectures across GCP services.

Q118.

What is App Engine dispatch.yaml?

Answer:

dispatch.yaml overrides App Engine's default URL routing. Define routing rules mapping URL patterns to services. Example: route `/api/*` requests to the 'api' service, `/admin/*` to the 'admin' service, all others to default service. Supports glob (* wildcard) patterns. Maximum 20 dispatch rules per application. Deploy: `gcloud app deploy dispatch.yaml`. Rules are evaluated in order; first match wins.

Q119.

What is App Engine Flexible Environment networking?

Answer:

Flexible environment VMs are deployed directly in your project's VPC network. You can: configure VPC network and subnet, apply network tags for firewall rule targeting, use internal IPs for private resource access, connect to Cloud SQL via private IP, and access Memorystore directly. Unlike Standard environment (which needs Serverless VPC Access), Flexible is natively a VPC resource with full VPC networking capabilities.

Q120.

When would you choose App Engine over Cloud Run?

Answer:

Choose App Engine when: team prefers no-container deployment (just push code), you want built-in cron jobs and task queue integration, your app fits supported runtimes, you have an existing App Engine deployment, or you need the simplest possible PaaS experience. Choose Cloud Run when: you use containers (custom runtimes, multi-stage builds), need WebSocket support, want longer timeouts, use gRPC, need more control over the runtime environment, or want container portability.

Section 5: Cloud Storage & Databases

30 questions with comprehensive, interview-focused answers.

Q121.

What is Cloud Storage and its storage classes?

Answer:

Cloud Storage is GCP's unlimited, globally accessible object storage with 99.999999999% (11 nines) durability. Classes: Standard (hot data, no minimum duration, best latency/throughput), Nearline (accessed ~monthly, 30-day minimum, lower storage cost), Coldline (accessed ~quarterly, 90-day minimum), Archive (accessed yearly, 365-day minimum, lowest storage cost, highest retrieval fee). Objects auto-transition between classes via Lifecycle rules.

Q122.

What is the difference between Cloud Storage and Persistent Disk?

Answer:

Cloud Storage: object storage accessed via HTTP(S) API or gsutil. Stores blobs (files) with metadata. Not mountable as filesystem natively (use gcsfuse). Globally accessible, unlimited scale, billed per GB stored + operations. Persistent Disk: block storage attached to VMs as a filesystem (ext4, XFS). Accessed like local disk with low latency. Zonal or regional. Required for VM boot disks and databases. Completely different use cases – not interchangeable.

Q123.

What are Cloud Storage bucket access controls?

Answer:

Two access systems: Uniform Bucket-Level Access (recommended): IAM policies apply uniformly to all objects in the bucket – no per-object ACLs allowed. Simpler, more secure, required for VPC Service Controls. Fine-grained: combines IAM (bucket level) and legacy ACLs (per-object) – for backward compatibility with S3-style per-object access. Enable uniform access: `gcloud storage buckets update gs://BUCKET --uniform-bucket-level-access`. Cannot revert after 90 days.

Q124.

What is Storage Transfer Service?

Answer:

Storage Transfer Service moves data into Cloud Storage from: Amazon S3 buckets (cross-cloud migration), Azure Blob Storage, HTTP/HTTPS file lists, other GCS buckets, and on-premises data (via Transfer Service agent pools). Features: scheduled recurring transfers, source filtering (prefix, modification date), post-transfer deletion, checksum validation. For large on-premises transfers, also consider Transfer Appliance (physical device shipped to Google).

Q125.

What are Signed URLs in Cloud Storage?

Answer:

Signed URLs provide time-limited access to a specific GCS object without requiring Google authentication. The URL includes a cryptographic signature (signed by a service account key or using V4 signing). Anyone with the URL can access the object within the expiry period (max 7 days for V4). Uses: allow app users to download files without Google accounts, generate temporary upload URLs for user content. Generate via gsutil signurl, Storage client libraries, or IAM Credentials API.

Q126.

What is Cloud SQL?

Answer:

Cloud SQL is GCP's fully managed relational database service. Supported engines: MySQL (5.7, 8.0), PostgreSQL (9.6–15), SQL Server (2017, 2019). Google manages: automated backups, failover, patching, storage auto-grow, monitoring. Scales up to 96 vCPUs and 624 GB RAM. Supports private IP (recommended), Cloud SQL Auth Proxy, and public IP with authorized networks. Not globally distributed — use Cloud Spanner for global ACID transactions.

Q127.

What is Cloud SQL High Availability?

Answer:

Cloud SQL HA uses a synchronous primary-standby setup across two zones within a region. Every write is committed to both primary and standby before acknowledgment. Failover is automatic (60–120 seconds). Enable HA when creating or editing an instance (adds ~2x cost). Configure: `--availability-type=REGIONAL` in `gcloud`. HA does not provide read scaling – add read replicas for read offloading. Point-in-Time Recovery requires HA or automated backups with binary logging.

Q128.

What is Cloud SQL Auth Proxy?

Answer:

Cloud SQL Auth Proxy is a local binary/sidecar that provides secure, authenticated connections to Cloud SQL without managing SSL certificates or whitelisting IP addresses. It authenticates using IAM credentials (service account with roles/cloudsql.client), encrypts all traffic, and listens on a local socket (127.0.0.1:PORT). Works from: local development, VMs, Cloud Run, GKE (as sidecar). Required when connecting to Cloud SQL via external IP without SSL client certificates.

Q129.

What is Cloud Spanner?

Answer:

Cloud Spanner is Google's proprietary, globally distributed, horizontally scalable relational database. Unique properties: SQL + ACID transactions at global scale, 99.999% SLA (five nines), TrueTime for external consistency, automatic sharding. Scales by adding nodes (each node adds ~2,000 QPS and 2 TB storage). Use cases: global financial systems, inventory management, gaming leaderboards, any workload needing both relational semantics and horizontal scale.

Q130.

What is the difference between Cloud Spanner and Cloud SQL?

Answer:

Cloud SQL: standard MySQL/PostgreSQL/SQL Server, vertical scaling only, single-region primary (with cross-region read replicas), standard RDBMS features, lower cost (\$0.017/vCPU/hour for MySQL). Cloud Spanner: proprietary globally distributed database, horizontal scaling (add nodes), 99.999% SLA with multi-region config, globally consistent transactions, higher cost (\$0.90/node/hour + storage). Use Cloud SQL for standard workloads; Spanner for globally distributed, mission-critical, high-throughput applications.

Q131.

What is Cloud Firestore?

Answer:

Firestore is a serverless NoSQL document database. Data model: Collections □ Documents □ Fields (including nested subcollections). Two modes: Native (real-time listeners, offline sync, mobile SDKs, multi-document ACID transactions) and Datastore (App Engine backward-compatible, entity groups, no real-time). Scales automatically from zero. Strong consistency for all reads. No infrastructure to manage. Free tier: 1 GB storage, 50K reads/day, 20K writes/day.

Q132.

What is Cloud Bigtable?

Answer:

Bigtable is Google's fully managed, massively scalable NoSQL wide-column database (open-sourced as HBase). Designed for: time-series data (IoT metrics, financial tick data), AdTech, analytics. Architecture: sorted key-value store with row key + column family + qualifier + timestamp. Scales to petabytes. Replication across clusters for HA. HBase API compatible. Best for: single-row operations, high-throughput sequential scans. Not for: complex queries, relational joins.

Q133.

What is the difference between Firestore and Bigtable?

Answer:

Firestore: document NoSQL, rich querying (compound indexes, range filters), real-time listeners, mobile-friendly SDKs, multi-document ACID, serverless, free tier, up to ~1M QPS. Bigtable: wide-column NoSQL, single-row operations only, no complex queries, petabyte scale, millions of reads/writes per second, ~\$0.65+/hour minimum. Choose Firestore for app backends with complex query needs; Bigtable for ultra-high-throughput analytics or time-series data.

Q134.

What is BigQuery and how does it differ from transactional databases?

Answer:

BigQuery is GCP's fully managed, serverless data warehouse for analytics (OLAP). Columnar storage enables scanning billions of rows in seconds. Not designed for OLTP: high-latency for point lookups, optimized for scans over large datasets. Pricing: \$5/TB queried (on-demand) or flat-rate slot purchases. Features: partitioning, clustering, materialized views, ML (BQML), geospatial analysis, BI Engine for sub-second queries. Use alongside Cloud SQL – SQL for operations, BigQuery for analytics.

Q135.

What is Memorystore?

Answer:

Memorystore is GCP's fully managed in-memory data store. Redis support: versions 4–7, up to 300 GB, HA with automatic failover, replica for read scaling, import/export RDB files. Memcached support: up to 5 TB distributed, auto-discovery. Use cases: session caching, rate limiting, leaderboards, pub/sub, ML feature caching. Accessible only within the same VPC region. Pricing per GB of provisioned capacity per hour.

Q136.

What is Cloud Storage Object Lifecycle Management?

Answer:

Lifecycle Management automatically transitions or deletes objects based on configurable rules evaluated daily. Conditions: age (days since creation), created before date, storage class matches, is live/noncurrent version, number of newer versions. Actions: transition to cheaper storage class, delete object. Example: transition to Nearline after 30 days, Coldline after 90 days, delete after 365 days. Critical for cost management of large-scale storage.

Q137.

What is Object Versioning in Cloud Storage?

Answer:

Enable versioning on a bucket to preserve previous object versions when objects are overwritten or deleted. Overwriting creates a noncurrent version with a generation number; deleting creates a delete marker. Noncurrent versions can be retrieved, promoted, or permanently deleted. Use with lifecycle rules: delete noncurrent versions after N days, keep only N most recent versions. Protects against accidental deletion but increases storage costs.

Q138.

What is PITR in Cloud SQL?

Answer:

Point-in-Time Recovery restores a Cloud SQL database to any second within the past 7 days (requires binary log retention for MySQL or WAL for PostgreSQL). Use: accidentally deleted rows, corruption from a bad migration. Process: clone the instance to a timestamp before the incident (`gcloud sql instances clone SOURCE TARGET --point-in-time='TIMESTAMP'`), verify data, then either promote the clone or export the recovered data. PITR requires automated backups to be enabled.

Q139.

What are Cloud Storage Retention Policies?

Answer:

Retention policies set a minimum retention period for objects in a bucket — objects cannot be deleted or modified until the period expires. Lock a retention policy to make it immutable (permanently enforced — WORM compliance). Used for: SEC Rule 17a-4, FINRA, HIPAA, and other regulations requiring immutable audit trails. Once locked, the retention policy duration can only be increased, never shortened or removed. Plan carefully before locking.

Q140.

What is Datastream?

Answer:

Datastream is GCP's serverless change data capture (CDC) and replication service. Replicates data changes from Oracle, MySQL, PostgreSQL, and SQL Server to BigQuery, Cloud Storage, Cloud Spanner, and Pub/Sub in real time (sub-minute latency). Uses: real-time analytics, event-driven architectures, database migration with minimal downtime. Fully managed — no infrastructure to provision. Handles initial backfill plus ongoing CDC automatically.

Q141.

What is the difference between OLTP and OLAP in GCP?

Answer:

OLTP (Online Transactional Processing): optimized for many small, concurrent read/write transactions with low latency. GCP OLTP options: Cloud SQL (MySQL, PostgreSQL, SQL Server), Cloud Spanner (global), Firestore (NoSQL), Bigtable (NoSQL). OLAP (Online Analytical Processing): complex queries over large historical datasets. GCP OLAP: BigQuery (petabyte-scale SQL analytics). Most production architectures use OLTP for operations and stream changes to BigQuery for analytics.

Q142.

What is Database Migration Service (DMS)?

Answer:

DMS is GCP's fully managed database migration service. Supported migrations: MySQL → Cloud SQL MySQL, PostgreSQL → Cloud SQL PostgreSQL/AlloyDB, Oracle → Cloud SQL PostgreSQL (heterogeneous with Schema Conversion Tool). Uses continuous replication (CDC) for minimal downtime – source database stays live during migration. Cutover at your chosen time. Free service (pay only for Cloud SQL/AlloyDB destination). Handles schema conversion, data migration, and validation.

Q143.

What is AlloyDB?

Answer:

AlloyDB is Google's fully managed, PostgreSQL-compatible database with 4x faster OLTP and 100x faster analytics than standard PostgreSQL. Architecture: separates compute from storage (Google-managed distributed storage layer). Built-in columnar engine accelerates analytical queries without ETL. AI-assisted vacuum management. 99.99% SLA with rapid failover (<60 seconds). Use for: demanding transactional workloads requiring high availability and mixed OLTP/OLAP access patterns.

Q144.

What is Cloud Storage FUSE (gcsfuse)?

Answer:

gcsfuse is an open-source adapter mounting Cloud Storage buckets as local Linux/macOS filesystems using FUSE. Useful for: ML training jobs reading datasets from GCS, batch scripts expecting POSIX file paths, legacy applications. Limitations: lower performance than local disk, no atomic multi-file operations, eventual consistency for metadata. Not suitable for: databases, high-IOPS workloads, or applications needing strong POSIX consistency semantics.

Q145.

What is the maximum object size in Cloud Storage?

Answer:

Maximum single object size: 5 TB. For uploads larger than 5 MB: use resumable uploads (automatically retry partial uploads on failure — enabled by default in gsutil and client libraries). For very large files: use the XML/JSON multipart API compatible with AWS S3's multipart upload. Minimum object size: 0 bytes (empty objects are valid and useful as directory markers or locks).

Q146.

What is Spanner's TrueTime API?

Answer:

TrueTime is Google's globally synchronized time API providing bounded clock uncertainty. GPS receivers and atomic clocks in Google's data centers synchronize time with uncertainty bounds (typically <10ms). Spanner uses TrueTime to assign globally monotonic commit timestamps: if T1 commits before T2 starts, T1's timestamp is always lower. This guarantees external consistency — reads after a commit see that commit, globally, without distributed coordination locks.

Q147.

What is Bigtable row key design?

Answer:

Row keys are sorted lexicographically — this defines data locality and access patterns. Design keys to: (1) Avoid hotspots — never use sequential IDs or timestamps as prefix (creates write hotspot on the latest tablet). (2) Achieve even load distribution — use hash prefixes. (3) Enable efficient scans — collocate frequently co-accessed rows. Best practice composite keys: reversed_timestamp#user_id (latest data first), or salted#entity_id (distributed writes with prefix lookups).

Q148.

What are Cloud SQL read replicas?

Answer:

Read replicas serve read traffic to offload the primary instance. Replication is asynchronous (replicas may lag primary by seconds). Types: In-region replica (same region, low replication lag), Cross-region replica (different region, useful for DR and geographic read locality), Cascaded replica (replica of a read replica — reduces load on primary for multi-tier replication). Replicas cannot accept writes. Promote to standalone: `gcloud sql instances promote-replica REPLICA_NAME`.

Q149.

What is Filestore?

Answer:

Filestore is GCP's managed NFS (v3) file storage service providing shared file system access for multiple compute resources simultaneously. Tiers: Basic HDD (low cost, 1-63.9 TB), Basic SSD (higher IOPS), Enterprise (99.99% SLA, snapshots, cross-region replication, 1-10 TB), High Scale SSD (extreme performance for HPC, up to 100 TB). Use cases: shared content repositories, EDA tool storage, CMS file systems, game assets, HPC scratch storage.

Q150.

What is the difference between regional and multi-regional Cloud Storage?

Answer:

Regional: data stored in a single GCP region (e.g., us-central1). Lowest latency for compute in the same region. Lower cost than multi-regional. 99.9% SLA. Best for: data used by VMs/GKE in one region. Multi-Regional: data geo-redundantly replicated across multiple regions within a continent (US, EU, ASIA). 99.95% SLA. Best for: globally accessed data, critical assets needing geo-redundancy. Use Dual-Region for specific two-region redundancy with a lower RPO than multi-regional.

Section 6: Networking — VPC, Subnets, Firewall

30 questions with comprehensive, interview-focused answers.

Q151.

What is a VPC in GCP?

Answer:

A Virtual Private Cloud (VPC) is a global, private, software-defined network in GCP. Unlike AWS (VPCs are regional), GCP VPCs span all regions globally. Subnets are regional resources within the VPC. Resources in different regions within the same VPC can communicate via internal IP addresses using Google's global network backbone without traversing the public internet. One default VPC per project; create up to 5 additional VPCs.

Q152.

What is the difference between Auto-mode and Custom-mode VPC?

Answer:

Auto-mode: one subnet per region automatically created using predefined /20 IP ranges from 10.128.0.0/9. Quick to start; default VPC uses auto-mode. Limitations: can't use custom IP ranges, IP ranges may conflict with on-premises networks. Custom-mode: no subnets created automatically; you define all subnets, regions, and CIDR ranges explicitly. Required for production, VPN/Interconnect with on-premises, and controlled IP address planning. Cannot convert auto to custom mode.

Q153.

What is VPC Peering?

Answer:

VPC Peering connects two VPC networks (same or different projects/organizations) allowing internal IP communication. Traffic stays on Google's network — no internet traversal, no bandwidth cost (standard GCP internal rates). Key limitations: non-transitive ($A \square B, B \square C$ does NOT mean $A \square C$), no overlapping IP ranges, maximum 25 peers per VPC, no peering with Shared VPC. Each side must create a peering separately; both must be active for connectivity.

Q154.

What is Shared VPC?

Answer:

Shared VPC designates one project as the host project whose VPC network is shared with service projects. Resources in service projects (VMs, GKE nodes) use subnets from the host VPC but remain in their own projects for billing and IAM. Benefits: centralized network management (single team owns firewall rules, routes), consistent security policy enforcement, simplified connectivity between projects. Requires Organization Admin to configure. Incompatible with VPC peering between host and service projects.

Q155.

What is Cloud NAT?

Answer:

Cloud NAT (Network Address Translation) is a fully managed, distributed service allowing private VMs (without external IPs) to initiate outbound internet connections (for software updates, calling external APIs, pulling containers). Provides Source NAT only — external systems cannot initiate connections to VMs. Configured per region per VPC network. No proxy VM required — runs as a distributed GCP managed service. Logs available in Cloud Logging for egress audit.

Q156.

What are GCP Firewall Rules?

Answer:

Firewall rules control ingress/egress traffic at the VM network interface level. Key attributes: Direction (ingress/egress), Priority (0-65534, lower = higher priority), Action (allow/deny), Target (all instances, network tag, or service account), Source/Destination (IP ranges, network tags, service accounts), Protocol/Port. Implied rules: lowest priority deny-all ingress, allow-all egress. Rules are stateful (return traffic automatically allowed).

Q157.

What is the difference between network tags and service account targets in firewall rules?

Answer:

Network tags: arbitrary strings applied to VM metadata; firewall rules reference these tags as targets or sources. Simple to configure but any user with instance admin can add tags — potentially bypassing firewall intent. Service account targets: firewall rules apply to VMs running as specific service accounts. More secure — only IAM admins can assign service accounts. Recommended for security-sensitive environments where developers shouldn't control their own firewall exemptions.

Q158.

What is Cloud Load Balancing?

Answer:

GCP Load Balancing types: Global External HTTP(S) LB (layer 7, anycast, URL routing, SSL termination, CDN integration, Cloud Armor). Regional External HTTP(S) LB (layer 7, regional, supports Envoy-based advanced traffic management). Global SSL Proxy / TCP Proxy (layer 4, global). Regional External TCP/UDP Network LB (layer 4, pass-through, preserves client IP). Regional Internal HTTP(S) LB (layer 7, private VPC). Regional Internal TCP/UDP LB (layer 4, private VPC). Choose based on protocol, scope, and whether traffic is internal or external.

Q159.

What is the difference between internal and external load balancers?

Answer:

External Load Balancers accept traffic from the internet (public IP or Google's Anycast infrastructure). Internal Load Balancers distribute traffic within a VPC from internal clients — accessible only via RFC1918 private IP addresses, traffic never leaves Google's network. Internal LBs: Internal HTTP(S) LB (layer 7, Envoy proxy) and Internal TCP/UDP LB (pass-through, preserves client IP). Use internal LBs for microservices, database proxies, and service-to-service communication.

Q160.

What is Cloud DNS?

Answer:

Cloud DNS is GCP's 100% uptime SLA, scalable, managed authoritative DNS service. Features: public zones (internet-facing DNS), private zones (internal VPC DNS resolution, not public), DNS peering (share private zones across VPCs), split-horizon DNS (different answers for internal vs external), DNSSEC support. Integrates with GKE (ExternalDNS), Cloud Domains, and supports programmatic management via API/Terraform.

Q161.

What are VPC Flow Logs?

Answer:

VPC Flow Logs capture sampled network traffic metadata for VM interfaces: source/destination IP, port, protocol, bytes, packets, and RTT latency. Configured per subnet (enable/disable, sampling rate 0.0–1.0). Logs go to Cloud Logging. Uses: network forensics, security analysis, troubleshooting connectivity issues, cost attribution, compliance auditing. Enable for all subnets in production VPCs. Export to BigQuery for large-scale analysis.

Q162.

What is Cloud Armor?

Answer:

Cloud Armor is GCP's DDoS protection and WAF. Integrates with Global and Regional HTTP(S) Load Balancers. Capabilities: L3/L4 volumetric DDoS protection (automatic), L7 WAF rules (OWASP CRS, custom CEL expressions), IP/CIDR allow/deny lists, geographic (country) access control, rate limiting per IP, Adaptive Protection (ML-based attack detection and automatic rule suggestions). Cloud Armor Managed Protection Plus includes advanced DDoS response support.

Q163.

What is BGP and how is it used in GCP?

Answer:

Border Gateway Protocol (BGP) is the internet's inter-domain routing protocol. GCP uses BGP for dynamic route exchange between GCP VPCs and on-premises networks via: Cloud Routers (required for HA VPN and Cloud Interconnect). Dynamic routing with BGP automatically propagates route changes without manual updates — add an on-premises subnet, BGP propagates it to GCP. Required for HA VPN (99.99% SLA requires BGP on both tunnels).

Q164.

What is a Cloud Router?

Answer:

Cloud Router is a managed BGP routing service enabling dynamic route exchange between GCP VPCs and external networks. Required for: HA VPN (both tunnels), Cloud Interconnect. Configuration: create Cloud Router per region, peer with on-premises router using BGP ASN and peer IPs, advertise custom routes. Global routing mode: a single Cloud Router in one region can learn and advertise routes for all regions in the VPC — useful for hub-and-spoke connectivity.

Q165.

What is HA VPN?

Answer:

HA VPN provides 99.99% SLA Google-to-peer VPN by using two GCP VPN gateways with two tunnels each (four tunnels total to peer gateways) over separate Google infrastructure. Both tunnels use BGP for dynamic routing. Maximum ~3 Gbps per tunnel (12 Gbps total). HA VPN replaces Classic VPN (99.9% SLA, static routes). Always use HA VPN for production on-premises connectivity. Configure on-premises peer with two VPN devices for true redundancy.

Q166.

What is Private Google Access?

Answer:

Private Google Access allows VM instances with only internal IP addresses to reach Google APIs and services (Cloud Storage, BigQuery, Cloud SQL Proxy, Container Registry) using GCP's private network rather than the internet. Enable per subnet: `gcloud compute networks subnets update SUBNET --region=REGION --enable-private-ip-google-access`. Required for private VMs in subnets without external IPs that need to call Google APIs.

Q167.

What is the difference between North-South and East-West traffic in GCP?

Answer:

North-South: traffic between the internet (or external networks) and GCP resources. Managed by: External Load Balancers, Cloud Armor (WAF/DDoS), Cloud NAT (outbound), Cloud Interconnect/HA VPN (on-premises). East-West: traffic between GCP resources within the network (VM-to-VM, pod-to-pod, microservice-to-microservice). Managed by: VPC firewall rules, Internal Load Balancers, Anthos Service Mesh. Secure both directions with appropriate controls.

Q168.

What is the default route in GCP VPC?

Answer:

The default route (0.0.0.0/0) points to the internet gateway, allowing VMs with external IPs or Cloud NAT to reach the internet. Removing the default route creates an air-gapped private VPC with no internet access. Custom routes can override it: route on-premises traffic (10.0.0.0/8) through HA VPN tunnel, route specific internet traffic (203.0.113.0/24) through an NFV appliance. Priority determines which route wins when multiple match.

Q169.

What is Packet Mirroring?

Answer:

Packet Mirroring clones traffic from specified VMs and forwards it to an IDS/monitoring collector without impacting the original traffic flow. Configure Packet Mirroring policies specifying: mirrored sources (by instance, subnet, or network tag), collector (Internal TCP/UDP Load Balancer frontend), and traffic filter (protocol, CIDR, direction). Used for: intrusion detection systems, network forensics, compliance monitoring. Collector receives full packet payload for deep inspection.

Q170.

What is Cloud CDN?

Answer:

Cloud CDN caches content at 130+ Google Edge PoPs globally, reducing latency for end users and reducing load on origin backends. Integrates with Global HTTP(S) Load Balancer. Cache modes: `USE_ORIGIN_HEADERS` (respect Cache-Control), `CACHE_ALL_STATIC` (auto-detect static content), `FORCE_CACHE_ALL` (cache everything). Features: cache invalidation, signed URLs/cookies for private content, negative caching, cache bypass. Reduces time-to-first-byte for global users by serving from the nearest PoP.

Q171.

What is Private Service Connect?

Answer:

Private Service Connect (PSC) allows VPC consumers to access managed services via internal IP endpoints within their VPC without public internet exposure. Access: Google APIs (BigQuery, Cloud Storage) via PSC endpoint with custom DNS, Cloud SQL via PSC ILB, third-party services published via PSC. More flexible than Private Google Access: supports custom DNS names, access from on-premises via VPN/Interconnect, and publishing your own services to other VPCs.

Q172.

What is the MTU in GCP VPC?

Answer:

GCP VPC default MTU is 1460 bytes (standard Ethernet 1500 minus 20-byte IP header minus 20-byte TCP header). Configure VPC MTU to: 1500 bytes (standard Ethernet – compatible with most protocols), or 8896 bytes (Jumbo Frames – for HPC and high-throughput storage workloads). Set MTU when creating the VPC. All instances in the VPC use the VPC MTU. Jumbo frames can significantly improve throughput for large data transfers within the same VPC.

Q173.

What are Alias IP Ranges?

Answer:

Alias IP Ranges assign additional internal IP addresses (from the subnet's primary or secondary range) to a VM's network interface. Uses: host multiple services on one VM (different IPs per service), container networking (each container/pod gets its own IP). GKE VPC-native clusters use alias IP ranges from a secondary subnet range to assign pod IPs – pods become first-class VPC citizens, routable throughout the VPC without NAT.

Q174.

What is a VPC-native cluster in GKE?

Answer:

VPC-native GKE clusters use alias IP ranges for pod IPs, making pods directly routable within the VPC. Benefits: pods accessible from on-premises via VPN/Interconnect, VPC peering supports pod IP communication, Private Google Access works for pods, compatible with VPC Service Controls. Required for: VPC peering with pod IPs, most modern GKE features. All new GKE clusters should use VPC-native mode. Enable at cluster creation (`--enable-ip-alias` flag or via Console).

Q175.

What is Network Intelligence Center?

Answer:

Network Intelligence Center is GCP's network observability and troubleshooting suite. Components: Network Topology (visualize VPC architecture, traffic flows, and latency between GCP resources), Connectivity Tests (simulate and test reachability between endpoints by analyzing firewall rules, routes, and NAT — without sending real traffic), Performance Dashboard (network latency and packet loss metrics), Firewall Insights (identify shadowed, overly permissive, or unused firewall rules).

Q176.

What is the difference between TCP Proxy and SSL Proxy load balancers?

Answer:

Both are global layer-4 load balancers terminating connections at Google's edge PoPs. SSL Proxy: terminates SSL/TLS, handles certificate management, re-encrypts or sends plain TCP to backends. Use for SSL-based non-HTTP protocols. TCP Proxy: terminates TCP connections (no SSL handling), proxies TCP to backends. Use for non-SSL TCP protocols (IMAP, SMTP, game server protocols, LDAP). Both provide global anycast IPs and DDoS protection. Neither provides URL-based routing — use HTTP(S) LB for that.

Q177.

What is a Forwarding Rule in GCP load balancing?

Answer:

A Forwarding Rule is the frontend configuration of a GCP load balancer. It defines: IP address (ephemeral or static), protocol, and port(s) that trigger the load balancer. Each forwarding rule maps to exactly one target: a Target HTTP/HTTPS Proxy (for HTTP LB), Target SSL/TCP Proxy, Target Pool (Network LB), or Backend Service. Global forwarding rules use global anycast IPs; regional forwarding rules use regional IPs.

Q178.

What are secondary IP ranges in GCP subnets?

Answer:

Secondary IP ranges are additional CIDR blocks attached to a subnet, separate from the primary CIDR range. They exist exclusively for alias IP addresses — VMs can use IPs from secondary ranges as alias IPs for containers. GKE VPC-native clusters require two secondary ranges per subnet: one for pod IPs (e.g., /14 CIDR for up to 1,000 nodes × 110 pods each), one for service cluster IPs (e.g., /20). Secondary ranges don't overlap with primary ranges or other subnets.

Q179.

What are Cloud NAT port allocation modes?

Answer:

Cloud NAT port allocation modes: Manual (you specify min ports per VM — predictable but may over/under-allocate), Automatic (Cloud NAT dynamically allocates ports based on active connections — more efficient but harder to predict). Ports per NAT IP: 64,512 usable. Each TCP/UDP connection uses one port. If a VM needs more connections than allocated ports, it fails. Monitor: Cloud Monitoring metrics `nat/dropped_sent_packets_count` indicates port exhaustion issues.

Q180.

What is Network Endpoint Groups (NEG) in GCP?

Answer:

Network Endpoint Groups define backends for load balancers at the individual endpoint level (IP:port or FQDN:port) rather than at the instance group level. Types: Zonal NEG (VM endpoints within a zone — fine-grained health checking), Internet NEG (external backend FQDN/IP for on-premises or third-party backends), Serverless NEG (route to Cloud Run, App Engine, Cloud Functions — enables HTTPS LB in front of serverless), Hybrid connectivity NEG (on-premises via HA VPN/Interconnect). NEGs enable sophisticated traffic control and multi-backend architectures.

Section 7: Kubernetes Engine (GKE)

30 questions with comprehensive, interview-focused answers.

Q181.

What is Google Kubernetes Engine (GKE)?

Answer:

GKE is GCP's managed Kubernetes service. Google manages the control plane (API server, etcd, scheduler, controller manager) — including availability, upgrades, and backups — at no charge. You manage worker nodes (Standard mode) or Google manages everything (Autopilot mode). GKE natively integrates with Cloud Load Balancing, Cloud Storage, Cloud SQL, Artifact Registry, Cloud Monitoring/Logging, IAM, and other GCP services.

Q182.

What is the difference between GKE Standard and GKE Autopilot?

Answer:

Standard: you configure node pools (machine type, node count, auto-scaling), manage node-level settings, full control over cluster configuration. More operational overhead. Bill per node regardless of pod resource usage. Autopilot: Google manages all node infrastructure — you only submit pods. Billing per pod resource requests (not node capacity). Google enforces security best practices automatically. Autopilot recommended for most new workloads to reduce operational burden and improve security posture.

Q183.

What is a node pool in GKE?

Answer:

A node pool is a group of nodes within a GKE cluster sharing identical configuration: machine type, OS image, disk size/type, labels, taints, metadata, and auto-scaling settings. A cluster starts with a default node pool. Create additional pools for: GPU-accelerated ML workloads (A100 nodes), different OS requirements (Windows nodes), workload isolation via taints (dedicated pools for specific teams or services). Each pool can have independent auto-scaling min/max settings.

Q184.

How does GKE cluster auto-scaling work?

Answer:

Cluster Autoscaler (CA) monitors for unschedulable pods (insufficient CPU/memory/GPU in current nodes) and provisions new nodes from the appropriate pool. It also identifies underutilized nodes whose pods can be rescheduled elsewhere and removes them. CA respects: PodDisruptionBudgets, node taints, pod affinity/anti-affinity, and min/max node pool counts. Scale-up is aggressive (fast); scale-down is cautious (waits 10 minutes before removing an underutilized node).

Q185.

What is a GKE private cluster?

Answer:

A private cluster's nodes have only internal (private) IP addresses — no external IPs. Control plane can be private-only (no public endpoint) or with an authorized networks-restricted public endpoint. Private nodes use Private Google Access for pulling images from Artifact Registry and calling GCP APIs. Use Cloud NAT for outbound internet access (e.g., pulling public container images). Required for security-conscious production workloads. Configure at cluster creation — cannot change afterward.

Q186.

What is Workload Identity in GKE?

Answer:

Workload Identity allows GKE pods to authenticate to GCP services as a specific GCP Service Account without key files. Setup: (1) Annotate Kubernetes Service Account (KSA): `kubectl annotate serviceaccount KSA iam.gke.io/gcp-service-account=GSA@PROJECT.iam.gserviceaccount.com`. (2) Grant Workload Identity User role to the KSA binding on the GSA. Pods using that KSA automatically receive GSA's IAM permissions via the GKE metadata server. Recommended over mounting SA key files.

Q187.

What is a GKE Ingress?

Answer:

Deploying a Kubernetes Ingress resource on GKE triggers creation of a Google Cloud HTTP(S) Load Balancer. Features: SSL termination (managed certificates via ManagedCertificate resource), path-based and host-based routing to different Services, global anycast IP, Cloud Armor integration, Cloud CDN support, Backend Config for health checks and session affinity. External Ingress (gce IngressClass) creates external LB; Internal Ingress (gce-internal) creates internal LB.

Q188.

What are node taints and tolerations in GKE?

Answer:

Taints mark nodes to repel pods that don't tolerate them. Format: KEY=VALUE:EFFECT. Effects: NoSchedule (don't schedule new pods), PreferNoSchedule (avoid if possible), NoExecute (evict running pods). Tolerations in pod spec override specific taints. Pattern: taint GPU nodes with gpu:NoSchedule, add toleration only to ML pods — ensures GPU nodes are used exclusively for ML workloads. Combined with node affinity for precise scheduling control.

Q189.

What is GKE Vertical Pod Autoscaler (VPA)?

Answer:

VPA recommends and optionally adjusts CPU/memory requests and limits for pods based on observed usage history. Modes: Off (recommendations only), Initial (apply at pod creation only), Auto (update running pods — causes restarts). Useful for: setting initial resource requests, rightsizing pods over time. Limitations: don't use VPA and HPA on the same CPU metric (conflicts). VPA and HPA can coexist when HPA scales on custom metrics (Pub/Sub depth, business metrics).

Q190.

What is Horizontal Pod Autoscaler (HPA)?

Answer:

HPA automatically adjusts the number of pod replicas based on observed metrics. Default metrics: CPU utilization (target percentage of requests). Custom metrics: memory, Pub/Sub undelivered messages, custom Cloud Monitoring metrics via Custom Metrics Adapter. HPA evaluates metrics every 15 seconds. Scale-up is fast; scale-down has a 5-minute cooldown to avoid thrashing. Requires proper resource requests to calculate utilization percentage accurately.

Q191.

What is a DaemonSet in Kubernetes?

Answer:

A DaemonSet ensures exactly one pod runs on every node (or selected nodes via nodeSelector). Automatically adds pods to newly joined nodes and removes them from departing nodes. Use cases: log collection agents (Fluentd, Filebeat), monitoring agents (Prometheus node-exporter), network plugins (Calico, Weave), security agents (Falco), and storage daemons. DaemonSets bypass normal scheduling — pods run even on tainted nodes unless toleration is absent.

Q192.

What is Container-Optimized OS (COS) in GKE?

Answer:

COS (Container-Optimized OS) is Google's purpose-built OS for running containers. Design: stateless (read-only root filesystem), minimal packages (small attack surface), automatically updated by Google, built-in kernel hardening (seccomp, AppArmor, gVisor support), and verified boot. Default GKE node OS. Alternative: Ubuntu for nodes requiring custom kernel modules, specific filesystem support, or packages not in COS. COS nodes cannot install arbitrary packages.

Q193.

What is GKE Binary Authorization?

Answer:

Binary Authorization enforces that only cryptographically attested (signed) container images can be deployed. Workflow: Cloud Build builds image → vulnerability scanner attests clean scan → Binary Authorization policy requires attestation before deployment. Policy modes: Allow all (off), Dry-run (log violations without blocking), Warn (allow with warning), Enforce (block unsigned images). Integrates with Artifact Registry for attestation storage. Prevents deployment of unauthorized or tampered images.

Q194.

What is the GKE control plane?

Answer:

The GKE control plane runs Kubernetes master components: API Server (handles all kubectl and API requests), etcd (stores cluster state), Scheduler (assigns pods to nodes), Controller Manager (runs reconciliation loops for Deployments, DaemonSets, etc.), Cloud Controller Manager (integrates with GCP APIs for load balancers, persistent volumes). Google manages the control plane: automatic upgrades, HA (3 etcd instances across zones), and health monitoring.

Q195.

What are GKE Release Channels?

Answer:

Release channels manage automatic GKE version upgrades. Rapid: latest features first, most frequent upgrades, shortest support window — good for testing new features. Regular: recommended for most users, balanced between new features and stability. Stable: mature versions, longest support, slowest upgrades — for conservative production workloads. Enrolling in a channel gives Google permission to upgrade your cluster within the channel's version policy. Manual/static versions require you to trigger upgrades.

Q196.

What is a Pod Disruption Budget (PDB)?

Answer:

PDBs limit voluntary disruptions to running pods, ensuring minimum availability during: node upgrades, cluster autoscaler scale-down, draining operations. Specify: minAvailable (minimum pods that must remain up, absolute count or percentage) or maxUnavailable (maximum pods that can be down). GKE cluster autoscaler and kubectl drain respect PDBs. Without PDB, rolling updates or node drains might take down too many replicas simultaneously, causing service degradation.

Q197.

What is GKE Config Connector?

Answer:

Config Connector is a GKE add-on enabling GCP resource management via Kubernetes CRDs (Custom Resource Definitions). Represent GCP resources (Cloud SQL, Cloud Storage, Pub/Sub) as Kubernetes objects — manage with kubectl and gitops. Config Connector reconciles the desired state declared in Kubernetes manifests with actual GCP state. Requires Workload Identity for the Config Connector KSA. Useful for teams wanting a single Kubernetes-native management plane.

Q198.

What is Anthos Service Mesh (ASM)?

Answer:

ASM is Google's managed Istio-based service mesh for GKE clusters. Features: automatic mTLS between all services (zero-config encryption), traffic management (retries, circuit breakers, canary deployments, fault injection), advanced observability (request traces in Cloud Trace, service-level metrics in Cloud Monitoring, topology visualization in Cloud Console), and AuthorizationPolicies for service-to-service access control. Managed ASM auto-upgrades the Istio control plane.

Q199.

What is GKE Sandbox (gVisor)?

Answer:

GKE Sandbox runs containers in a user-space kernel sandbox using Google's gVisor runtime. The gVisor Sentry intercepts and handles syscalls from containers, preventing direct access to the Linux host kernel. Benefits: strong isolation for multi-tenant or untrusted workloads, container escape exploits hit gVisor rather than host kernel. Trade-offs: slight performance overhead for syscall-intensive workloads, some syscalls not fully supported. Enable with `--sandbox-type=gvisor` on node pool.

Q200.

What is node auto-provisioning in GKE?

Answer:

Node Auto-Provisioning (NAP) extends Cluster Autoscaler to automatically create new node pools when pending pods require resources (GPU types, specific machine types, larger resource amounts) not available in existing pools. NAP also deletes pools when no longer needed. Eliminates manual node pool pre-provisioning – cluster dynamically adapts to any workload shape. Configure resource limits (max total CPUs, memory, GPU count) to bound NAP expansion.

Q201.

How do you upgrade a GKE cluster?

Answer:

Step 1: Upgrade control plane: `gcloud container clusters upgrade CLUSTER_NAME --master --cluster-version=VERSION`. Step 2: Upgrade node pools: `gcloud container clusters upgrade CLUSTER_NAME --node-pool=POOL_NAME`. Use surge upgrades (maxSurge: add nodes before removing old ones) to minimize disruption. In release channels, control plane upgrades are automatic; node upgrades can be automatic or manual. Blue-green node pool upgrades: create new pool, migrate workloads, delete old pool – maximizes control.

Q202.

What are GKE node images?

Answer:

Node image options: COS (Container-Optimized OS, containerd runtime, default, recommended), COS with Docker (deprecated), Ubuntu (containerd or Docker, supports custom kernel modules), Ubuntu with Docker (deprecated), Windows Server (for Windows containers). All new clusters use containerd runtime (Docker shim removed in Kubernetes 1.24+). COS is the most secure and maintained; use Ubuntu when COS lacks specific kernel features needed by workloads.

Q203.

What are common kubectl commands for troubleshooting?

Answer:

`kubectl get pods -n NAMESPACE -o wide` (list pods with node assignment).
`kubectl describe pod POD_NAME` (events, container states, resource allocation).
`kubectl logs POD_NAME -c CONTAINER --previous` (logs from crashed container).
`kubectl exec -it POD_NAME -- /bin/bash` (shell access).
`kubectl top pods / nodes` (real-time resource usage).
`kubectl get events --sort-by=.lastTimestamp` (cluster events in order).
`kubectl rollout status deployment/DEPLOY` (track rollout progress).

Q204.

What is a StatefulSet in GKE?

Answer:

StatefulSets manage stateful applications where pod identity matters: stable network identity (pod-0, pod-1, pod-2 — predictable hostnames via headless service), ordered pod creation/scaling/deletion, stable persistent storage (each pod gets its own PVC, retained on pod rescheduling). Use for: distributed databases (Cassandra, MongoDB, MySQL with replication), message brokers (Kafka), and any application where pods are not interchangeable. Use Deployments for stateless applications.

Q205.

What is GKE Dataplane V2?

Answer:

GKE Dataplane V2 replaces iptables with eBPF (extended Berkeley Packet Filter) for packet processing. Benefits: network policy enforcement in microseconds (iptables takes minutes to update for large clusters), network policy logging (which flows were allowed/denied), lower CPU overhead for network operations. Also provides the foundation for GKE Cilium-based advanced networking features. Enable at cluster creation (immutable setting). Recommended for all new clusters with complex network policies.

Q206.

How do you configure resource requests and limits in Kubernetes?

Answer:

In pod spec under `containers[].resources`: requests (scheduling minimum — node must have this allocatable) and limits (maximum — CPU throttled, memory OOMKilled at limit). Example: `requests: {cpu: '100m', memory: '128Mi'}, limits: {cpu: '500m', memory: '512Mi'}`. Best practice: always set requests for scheduler accuracy. Set limits 2-3x requests for headroom. Use VPA to discover appropriate values. Namespaces can set `LimitRanges` (per-container defaults) and `ResourceQuotas` (namespace totals).

Q207.

What is flexible pod scheduling in GKE?

Answer:

Kubernetes provides multiple scheduling constructs: `nodeSelector` (schedule on nodes with exact label match — simplest), `Node Affinity` (required or preferred rules with richer label expressions), `Pod Affinity` (co-locate pods with other pods sharing labels — for performance), `Pod Anti-Affinity` (spread pods away from each other — for HA across nodes/zones), `Topology Spread Constraints` (evenly distribute pods across topology domains like zones or nodes — preferred over anti-affinity for even spreading).

Q208.

What are GKE Spot Nodes?

Answer:

GKE Spot Nodes use Compute Engine Spot VMs for node pools at up to 91% discount. GCP can preempt Spot nodes at any time with 30-second shutdown signal. GKE automatically evicts pods from preempted nodes and reschedules them on available nodes. Best for: batch jobs, CI/CD pipeline agents, ML training, and non-critical stateless workloads. Mix Spot and standard node pools: batch jobs on Spot, production services on standard, using node selectors or taints.

Q209.

What is the difference between ClusterIP, NodePort, and LoadBalancer services?

Answer:

ClusterIP: virtual IP accessible only within cluster, stable even as pods change. Default for internal microservice communication. NodePort: opens a port (30000-32767) on every node's IP — accessible externally but not production-ready (exposes node IPs, no load balancing). LoadBalancer: provisions a GCP Cloud Load Balancer with external IP, routes to pods. For production: use Ingress (HTTP/S with URL routing, SSL, Cloud Armor) or LoadBalancer (TCP/UDP). Internal LoadBalancer: set annotation `cloud.google.com/load-balancer-type: Internal`.

Q210.

What is GKE Gateway API?

Answer:

GKE Gateway API is the successor to Kubernetes Ingress, providing richer traffic management using Gateway, HTTPRoute, TCPRoute, and GRPCRoute resources. Benefits over Ingress: role-based splitting (infra team owns Gateway, dev team owns HTTPRoute), multi-cluster routing, traffic weighting for canary deployments, header-based routing, retry policies, and timeout configuration in a Kubernetes-native way. GKE Gateway integrates with Google Cloud Load Balancing (Global and Regional). Recommended for new deployments requiring advanced routing.

Section 8: Monitoring & Logging

30 questions with comprehensive, interview-focused answers.

Q211.

What is Google Cloud's operations suite?

Answer:

Cloud Operations Suite (formerly Stackdriver) is GCP's integrated observability platform. Components: Cloud Monitoring (metrics collection, dashboards, alerting, SLOs, uptime checks), Cloud Logging (log ingestion, storage, routing, analysis), Cloud Trace (distributed request tracing), Cloud Profiler (continuous production CPU/memory profiling), Error Reporting (error aggregation and alerting). Works for GCP, AWS, on-premises, and hybrid environments.

Q212.

What can Cloud Monitoring monitor?

Answer:

Cloud Monitoring collects: GCP built-in metrics (automatically from all GCP services), agent metrics (from Ops Agent on VMs — memory, disk, application-specific), external metrics (from AWS CloudWatch, Prometheus, custom APIs), and user-defined custom metrics. Creates: dashboards, alerting policies (threshold, absence, ratio, forecast conditions), uptime checks (HTTP/TCP probes), and SLOs with error budget tracking. Supports MQL (Monitoring Query Language) and PromQL.

Q213.

What is the Ops Agent?

Answer:

Ops Agent is the modern unified agent for Compute Engine VMs replacing the legacy Monitoring Agent (collectd) and Logging Agent (fluentd). It collects: system metrics (CPU, memory, disk, network, process-level), application metrics and logs from 200+ integrations (MySQL, Apache, Redis, JVM, etc.), and custom log files. Configured via YAML at `/etc/google-cloud-ops-agent/config.yaml`. Sends data simultaneously to Cloud Monitoring and Cloud Logging.

Q214.

What is Cloud Logging?

Answer:

Cloud Logging is GCP's fully managed log management service. Ingests logs from: GCP services (automatic), applications (via client libraries or Logging API), VMs (via Ops Agent), and on-premises/AWS (via agent). Log entries structure: timestamp, resource (type + labels), severity, log name, textPayload or jsonPayload (structured), httpRequest, trace ID. Query with Logging Query Language (LQL) in Logs Explorer. Aggregate and export via log sinks.

Q215.

What are log bucket types in Cloud Logging?

Answer:

_Required bucket: stores Admin Activity and System Event audit logs. 400-day retention, cannot be disabled, no charge, cannot be modified. _Default bucket: stores all other logs by default. 30-day retention (configurable). Charged after free tier. Custom log buckets: user-defined names, 1-3650 day retention, specific regions for data residency, log views for access control within a bucket. Use Log Analytics feature for SQL queries on logs in BigQuery-integrated buckets.

Q216.

What is a log sink in Cloud Logging?

Answer:

A log sink routes matching log entries to a destination. The log router evaluates all sinks for each log entry and exports matching entries. Destinations: Cloud Storage (archival at low cost), BigQuery dataset (long-term analytics, SQL queries), Pub/Sub topic (real-time streaming to SIEM, third-party tools), Cloud Logging bucket (centralized logging from multiple projects). Sinks can be project-level, folder-level, or org-level (with --include-children to capture all child logs).

Q217.

What is a metric-based alerting policy?

Answer:

An alerting policy defines when to alert and who to notify. Components: Condition (metric threshold — e.g., CPU > 80% for 5 minutes; absence — metric stops reporting; ratio — error rate > 1%), notification channels (email, PagerDuty, Slack, SMS, Pub/Sub webhook), duration (how long the condition must be true before alerting), and documentation (description with alert context and runbook links). Multiple conditions per policy combined with AND/OR logic.

Q218.

What are uptime checks in Cloud Monitoring?

Answer:

Uptime checks probe endpoints from multiple Google regions every 1–5 minutes. Types: HTTP/HTTPS (verify response code, optional content match), TCP (verify port is open). Configure: request path, port, authentication headers, response content check. Multi-region: check from all selected regions simultaneously; alert when N of M regions fail. Results visible in Cloud Monitoring uptime check dashboard. Create alerting policy on the check to receive notifications on failure.

Q219.

What is Cloud Trace?

Answer:

Cloud Trace collects distributed tracing data from applications to show how requests flow through services and where latency occurs. Auto-instrumented for App Engine, GKE Ingress, and Cloud Run. For custom instrumentation: use OpenTelemetry (preferred), Cloud Trace client libraries, or Zipkin/Jaeger compatibility. Features: latency distribution for any URL, per-service breakdown, trace waterfall view, and automatic latency regression detection. Essential for diagnosing performance issues in microservices architectures.

Q220.

What is Cloud Profiler?

Answer:

Cloud Profiler continuously samples CPU time and heap memory in production applications with <1% overhead. Supports: Python, Java, Go, Node.js. Visualizes data as flame graphs – wider bars mean more time/memory spent. Helps identify: CPU hotspots (functions consuming excessive compute), memory leaks (growing heap over time), GC pressure (garbage collection overhead). No need to reproduce issues in test environments – profiles real production traffic patterns.

Q221.

What are log-based metrics?

Answer:

Log-based metrics extract quantitative data from log entries for use in Cloud Monitoring. Counter metric: counts log entries matching a filter per unit time (e.g., count ERROR logs per minute). Distribution metric: extracts a numeric value from matching entries for histogram/percentile analysis (e.g., extract response_time from access logs). Log-based metrics appear in Cloud Monitoring alongside other metrics – use in dashboards, alerting, and SLO definitions.

Q222.

What is the Logs Explorer?

Answer:

Logs Explorer is the Cloud Logging web UI for querying and analyzing logs. Features: Logging Query Language (LQL) with field autocomplete, Log Fields panel (interactive filter by resource type, project, severity, log name), histogram showing log volume over time, streaming mode (live tail), saved queries, direct linking to Cloud Trace for correlated traces. Advanced: log analysis queries (SQL) for Log Analytics-enabled buckets. Access via Console ▢ Logging ▢ Logs Explorer.

Q223.

What is Error Reporting?

Answer:

Error Reporting automatically groups application exceptions from Cloud Logging based on stack trace similarity. Surfaces: error count trends over time, first/last seen timestamps, sample payloads, affected user count, and link to relevant log entries. Supports: App Engine, Cloud Functions, Cloud Run, GKE, Compute Engine. Alerts on first occurrence of new error types. Integrates with PagerDuty and Slack. Enable programmatic error reporting via client libraries for better stack trace grouping.

Q224.

What is a custom metric in Cloud Monitoring?

Answer:

Custom metrics extend monitoring beyond GCP built-ins. Create via: Cloud Monitoring API (write time series via `TimeSeries.create`), client libraries (Python: `monitoring_v3`, Java, Go), or Ops Agent (scrape custom prometheus endpoints). Define metric descriptor: type (`custom.googleapis.com/your/metric`), value type (`INT64`, `DOUBLE`, `BOOL`, `DISTRIBUTION`), metric kind (`GAUGE`, `DELTA`, `CUMULATIVE`), and labels for dimensions. Use in alerting policies and dashboards identically to built-in metrics.

Q225.

What is Metric Explorer in Cloud Monitoring?

Answer:

Metric Explorer provides an ad-hoc charting interface for exploring any metric without creating a permanent dashboard. Select: metric type (filter by resource type), add filters (resource labels, metric labels), choose aggregation (sum, mean, max, min, percentile), apply functions (delta, rate, align), overlay multiple metrics. Build complex charts, compare time ranges, and identify correlations. Save to a dashboard when satisfied with the query. Supports both MQL and PromQL query modes.

Q226.

What is Service Monitoring and SLOs?

Answer:

Service Monitoring creates Service Level Objectives (SLOs) within Cloud Monitoring to track reliability. Define SLI (Service Level Indicator): request-based (e.g., availability = $\text{good_requests} / \text{total_requests}$, latency = % of requests under 500ms) or window-based. Set SLO target (e.g., 99.9% availability over 30 days). Cloud Monitoring tracks error budget consumption and burn rate. Alert when burn rate is too high (fast error budget consumption predicts SLO breach). Required for proper SRE practice.

Q227.

What is the difference between Cloud Logging and Cloud Audit Logs?

Answer:

Cloud Logging is the general log management service for all log types (application logs, system logs, GCP service logs). Cloud Audit Logs are a specific log stream within Cloud Logging recording GCP API calls: Admin Activity (always-on, who changed IAM, created VMs), Data Access (optional, who read data), System Event (GCP-generated events like live migration), and Policy Denied (access denied by VPC Service Controls). Audit Logs serve compliance and forensics; general logs serve debugging and monitoring.

Q228.

How do you extend log retention in Cloud Logging?

Answer:

Options: (1) Create a custom log bucket with up to 3,650 days (10 years) retention. Configure log sink to route logs to this bucket. (2) Export to Cloud Storage for indefinite retention at low cost (via a GCS log sink). (3) Export to BigQuery for analytics on historical logs (via BigQuery log sink). (4) For compliance: lock a Cloud Storage bucket with a retention policy for WORM-compliant log archival. Default _Default bucket retention is 30 days — export before logs expire.

Q229.

What are log exclusion filters?

Answer:

Exclusion filters prevent specified log entries from being ingested into Cloud Logging, reducing cost and noise. Configure in the Log Router: Logging ▢ Log Router ▢ Create Exclusion. Use LQL filter to match entries to exclude: exclude noisy debug logs (`severity=DEBUG AND resource.type='k8s_container'`), sample high-volume logs (`random_sample(0.01)` excludes 99%), exclude specific containerized apps. Warning: excluded logs are permanently dropped — cannot be recovered. Consider sampling over exclusion for audit-critical sources.

Q230.

What are Cloud Monitoring dashboards?

Answer:

Dashboards visualize metrics, logs, and alerts in a customizable UI. Widget types: metric charts (line, stacked area, bar, heatmap, scatter), incident list, uptime check summary, error reporting, log panels, and text annotations. Create via Console, API, or Terraform (`google_monitoring_dashboard`). Share across the project. GKE Dashboard, Cloud Run Dashboard, and VM Dashboard are pre-built. Pin frequently used dashboards. Embed metric charts in Google Docs/Slides via dashboard shareable links.

Q231.

What is alerting latency in Cloud Monitoring?

Answer:

Alerting latency is the time between a condition occurring and notification being sent. Components: metric ingestion lag (typically 30–90 seconds for GCP metrics), alignment period (how often metrics are aggregated), and condition duration (how long condition must persist before alerting). Total effective alerting time = ingestion lag + alignment period + condition duration. For faster alerting: reduce condition duration, use shorter alignment periods. Trade-off: lower duration increases false positive rate.

Q232.

What are the three pillars of observability in GCP?

Answer:

GCP's observability follows the standard three pillars: (1) Metrics (Cloud Monitoring): quantitative time-series measurements of system behavior — CPU%, request rates, error counts, latency percentiles. (2) Logs (Cloud Logging): discrete event records with context — application errors, audit trails, access logs. (3) Traces (Cloud Trace): request flow through distributed services showing end-to-end and per-service latency. Adding Profiling (Cloud Profiler) as a fourth pillar enables code-level performance analysis.

Q233.

What is Assured Workloads?

Answer:

Assured Workloads configures GCP environments to meet specific compliance frameworks including FedRAMP Moderate/High, DoD IL2/IL4/IL5, ITAR, CJIS, HIPAA, and others. It restricts: data location (only specific regions), personnel access (only US persons for government workloads), and available GCP products (only compliant services). Organizations enforce policies. Compliance monitoring in SCC detects violations. Required for US government contracts.

Q234.

What is Synthetic Monitoring?

Answer:

Synthetic Monitoring (Cloud Monitoring Synthetic Tests) runs scripted browser-based tests on a schedule to proactively detect issues before users report them. Uses Puppeteer/Playwright scripts to: simulate user journeys (login, add to cart, checkout), test multi-step API flows, verify third-party integrations. Runs from multiple GCP regions. Results include pass/fail status and response time. Set alerting policies on synthetic test results just like uptime checks.

Q235.

What is the Logging Query Language (LQL)?

Answer:

LQL filters log entries in Cloud Logging. Key syntax: `resource.type='gce_instance'` (resource type filter), `severity>=ERROR` (severity filter), `jsonPayload.statusCode=500` (structured log field), `textPayload:('connection refused')` (text search with :), `timestamp>='2024-01-15T00:00:00Z'` (time filter), `labels.pod_name='my-pod'` (label filter), AND / OR / NOT (logical operators). Use `=` for exact match, `:` for substring/contains. Field paths use dot notation for nested JSON fields.

Q236.

What is the difference between GAUGE, DELTA, and CUMULATIVE metrics?

Answer:

GAUGE: snapshot value at a point in time (e.g., current memory usage in bytes, current connected sessions). Aggregate by averaging or taking last value. DELTA: change over the alignment period (e.g., request count since last sample). Aggregate by summing. CUMULATIVE: monotonically increasing total since a start time (e.g., total bytes written since boot). Aggregate by `rate()` to convert to per-second. Understanding metric kind is essential: summing gauges is wrong; averaging deltas is wrong.

Q237.

How do you monitor GKE workloads effectively?

Answer:

GKE monitoring stack: (1) Enable GKE Monitoring features (System, API Server, Scheduler metrics). (2) Use GKE Workloads dashboard in Cloud Monitoring — pods, containers, CPU/memory. (3) Install Managed Service for Prometheus for application metrics without running Prometheus servers. (4) Configure HPA with custom metrics from Pub/Sub or Cloud Monitoring. (5) Set up log-based alerts for ERROR-level container logs. (6) Use Cloud Trace with OpenTelemetry for request tracing. (7) Define SLOs on key service endpoints.

Q238.

What is Managed Service for Prometheus (GMP)?

Answer:

Managed Service for Prometheus is GCP's fully managed Prometheus-compatible monitoring service. Ingest metrics: no changes to Prometheus-instrumented applications, just configure PodMonitoring CRDs to define scrape targets. GMP stores metrics in Google's managed backend at petabyte scale. Query with standard PromQL. Create alerts, dashboards, and SLOs in Cloud Monitoring using PromQL. Eliminates: running Prometheus servers, Thanos for long-term storage, Alertmanager. Pay per sample ingested.

Q239.

What is a Monitoring Workspace?

Answer:

A Cloud Monitoring Workspace (now called a scoping project) is a project that serves as the central view for monitoring data from multiple GCP projects. Add monitored projects to the workspace to view their metrics, uptime checks, alerting policies, and dashboards in a unified interface. Useful for centralized operations teams managing 50+ projects — single-pane-of-glass view without switching project contexts. Workspace configuration (dashboards, alert policies) is stored in the scoping project.

Q240.

What is the difference between push and pull subscriptions in Pub/Sub and how does it affect monitoring?

Answer:

Pull subscriptions: subscribers call Pub/Sub API to receive messages — subscriber controls rate, can process offline, monitor with `oldest_unacked_message_age` (latency) and `num_undelivered_messages` (backlog). Push subscriptions: Pub/Sub delivers messages to a HTTPS endpoint — simpler for serverless (Cloud Run, App Engine), monitor with `delivery_attempt_count` (retry storms) and `subscription/num_undelivered_messages`. Alert when `oldest_unacked_message_age` exceeds SLA threshold — indicates consumer lag or processing failure. Use pull for high-throughput; push for low-throughput event-driven architectures.

Section 9: Security & Best Practices

30 questions with comprehensive, interview-focused answers.

Q241.

What are the layers of security in GCP?

Answer:

GCP implements defense-in-depth: (1) Physical Security: custom-designed data centers, biometric access, 24/7 security staff, hardware destruction. (2) Hardware Security: Titan security chips providing hardware root of trust, custom-designed network equipment. (3) Network Security: encryption in transit (TLS/ALTS), private GCP backbone, DDoS protection. (4) Identity: IAM, MFA, BeyondCorp Zero Trust. (5) Application: code auditing, vulnerability scanning. (6) Data: AES-256 encryption at rest, CMEK, DLP API.

Q242.

What is BeyondCorp Enterprise?

Answer:

BeyondCorp Enterprise is Google's Zero Trust network access solution. Principle: no implicit trust based on network location — verify every access request explicitly. Components: Cloud Identity-Aware Proxy (application-level access based on identity + context), Access Context Manager (define access levels: device health, IP range, user identity), and Endpoint Verification (check device posture). Enables secure remote work without traditional VPN — employees access resources from any network with appropriate verification.

Q243.

What is the Cloud DLP API?

Answer:

Cloud Data Loss Prevention (DLP) API discovers, classifies, and de-identifies sensitive data (PII, PHI, PCI data). InfoTypes: pre-defined (US SSN, credit card, email, IP address) and custom patterns. Actions: Inspect (find sensitive data in text, files, BigQuery, Cloud Storage), De-identify (redact, mask, tokenize, encrypt, date-shift), Re-identify (restore with appropriate keys). Integrates with Dataflow for streaming pipelines. Essential for GDPR, HIPAA, and PCI DSS data governance.

Q244.

What is Cloud KMS?

Answer:

Cloud Key Management Service manages cryptographic keys. Key hierarchy: KeyRing (container per region) □ CryptoKey (key with rotation schedule) □ CryptoKeyVersion (actual key material). Operations: encrypt/decrypt data, sign/verify, generate random bytes. IAM controls access: roles/cloudkms.cryptoKeyEncrypterDecrypter for data operations. Key material never leaves KMS unencrypted. FIPS 140-2 Level 1 (software) and Level 3 (HSM) certified. Supports: AES-256-GCM, RSA-2048/4096, EC-P256/P384.

Q245.

What is Secret Manager?

Answer:

Secret Manager securely stores sensitive configuration: API keys, passwords, certificates, connection strings. Secrets are versioned (rotate without code changes), encrypted at rest with CMEK support, and access-controlled via IAM (roles/secretmanager.secretAccessor). Features: automatic rotation with Pub/Sub notification triggers, regional replication for data residency, audit logging in Cloud Audit Logs, and integration with Cloud Functions, Cloud Run, GKE, and Compute Engine. Eliminates secrets in environment variables, code, or config files.

Q246.

What is Security Command Center (SCC)?

Answer:

SCC is GCP's unified security and risk management platform. Standard tier: asset inventory, basic vulnerability findings. Premium tier: Security Health Analytics (continuous misconfiguration scanning – 150+ detectors), Event Threat Detection (ML-based threat detection from logs and GCP activity), Container Threat Detection (runtime container attacks), VM Threat Detection (memory-based malware detection), Web Security Scanner, and compliance dashboards (CIS, PCI DSS, HIPAA, NIST 800-53, ISO 27001). Premium is required for meaningful security posture management.

Q247.

What is the CIS GCP Foundation Benchmark?

Answer:

CIS (Center for Internet Security) GCP Foundation Benchmark provides 70+ security configuration recommendations across: IAM (enforce MFA, remove primitive roles, rotate SA keys <90 days), Logging (enable audit logs for all services, export logs), Networking (no default VPC, restrict firewall rules, no SSH/RDP from 0.0.0.0/0), VM (shielded VMs, OS Login), Storage (no public buckets, versioning, logging), KMS (key rotation), BigQuery (no public tables). SCC Premium auto-checks most controls.

Q248.

What is DNSSEC?

Answer:

DNSSEC (DNS Security Extensions) adds cryptographic signatures to DNS records to prevent DNS spoofing and man-in-the-middle attacks. Enable on Cloud DNS zones: Google generates DNSKEY (public key) and RRSIG (signature) records for every record set in the zone. Register DS (Delegation Signer) records with your domain registrar to complete the chain of trust from root to your zone. Required for US government domains and recommended for any public-facing service handling sensitive data.

Q249.

What is data minimization in GCP?

Answer:

Data minimization (GDPR Article 5): collect and retain only data necessary for the stated purpose. GCP implementation: Cloud DLP to scan and remove unnecessary PII, BigQuery table expiration policies, Cloud Storage lifecycle rules to delete old data, Logging exclusion filters to avoid logging sensitive fields, separate data classifications with different retention policies, restrict data collection at the application level, and use tokenization (DLP) to replace PII with pseudonymous tokens in analytics pipelines.

Q250.

What is Cloud Armor security policy?

Answer:

Cloud Armor security policies are ordered rule sets evaluated against incoming HTTP(S) traffic. Rule components: priority (evaluated lowest first), match condition (IP/CIDR, region, custom CEL expression matching headers/URL/body), action (allow/deny/rate-limit/redirect), and preview mode (log without enforcing). Use for: IP blocklists/allowlists (partner whitelisting, blocking scrapers), geographic restrictions (allow only US/EU), OWASP WAF rules (SQLi, XSS, RFI/LFI protection), and rate limiting per IP or region.

Q251.

What is GCP's encryption at rest?

Answer:

All GCP data at rest is encrypted by default using AES-256 (data encryption key, DEK) wrapped with AES-256 key encryption key (KEK) — envelope encryption. Encryption tiers: Google-managed encryption (default, free), Customer-Managed Encryption Keys (CMEK) via Cloud KMS (you control key rotation, revocation — if you revoke a key, Google cannot decrypt your data), Customer-Supplied Encryption Keys (CSEK) for Compute Engine/Cloud Storage (key never stored by Google), and External Key Manager (keys hosted outside GCP).

Q252.

What is GCP's encryption in transit?

Answer:

GCP encrypts all data in transit: Between users and GCP services: TLS 1.2 minimum (TLS 1.3 supported), HTTP Strict Transport Security (HSTS) on all endpoints. Between Google data centers: Application Layer Transport Security (ALTS) — Google's equivalent of mutual TLS using Handshake Protocol and record protocol for integrity. Cloud Load Balancers terminate TLS and support modern cipher suites (ECDHE, ChaCha20-Poly1305). HA VPN adds IPsec encryption for on-premises connections.

Q253.

What are Titan security chips?

Answer:

Titan is Google's custom-designed security chip embedded in GCP servers and network equipment. Functions: (1) Hardware root of trust — establishes verified boot sequence ensuring only authentic, unmodified firmware runs; (2) Hardware TPM — stores cryptographic keys and performs attestation; (3) Host authentication — Titan chip identity verified before server joins Google's production network. This protects GCP infrastructure against supply chain attacks and firmware tampering, ensuring hardware integrity.

Q254.

What is Security Health Analytics in SCC?

Answer:

Security Health Analytics (SHA) in SCC continuously scans GCP resource configurations for security misconfigurations. 150+ detectors covering: IAM (allUsers on IAM policies, primitive roles, SA keys >90 days old, public Cloud Functions), Network (overly permissive firewall rules, public RDP/SSH access, no VPC flow logs), Storage (public buckets, no versioning on logging buckets), Compute (no OS Login, shielded VM disabled, public VM without load balancer). Findings appear in SCC with severity and remediation guidance.

Q255.

What is OS Login in Compute Engine?

Answer:

OS Login centralizes SSH access control for Compute Engine VMs using IAM instead of managing SSH keys in instance metadata. Grant access: roles/compute.osLogin (normal user, no sudo) or roles/compute.osAdminLogin (sudo access). Benefits: (1) Revoking IAM role immediately revokes SSH access — no need to remove keys from 100+ VMs; (2) User accounts on VMs are automatically created based on Google identity; (3) All SSH access logged in Cloud Audit Logs with user identity. Enable via metadata: enable-oslogin=true.

Q256.

What are security best practices for Compute Engine?

Answer:

Enable OS Login (disable project-wide SSH keys), use Shielded VMs (Secure Boot + vTPM + Integrity Monitoring), assign dedicated minimal-privilege service account (not default compute SA), disable external IP if not needed (private VMs + Cloud NAT), enable VPC Flow Logs on all subnets, use VM Manager for automated OS patching, encrypt all disks with CMEK, apply principle of least privilege for firewall rules (deny-all default, allow specific), enable serial port access logging, and use Sole-Tenant Nodes for compliance isolation.

Q257.

What is container image security for GKE?

Answer:

Security-in-depth for container images: use minimal base images (distroless, alpine, scratch – reduce CVE surface), scan images for vulnerabilities with Artifact Registry scanning (built-in via Container Analysis API), enforce Binary Authorization (only signed, attested images deploy), configure pod security standards (runAsNonRoot: true, readOnlyRootFilesystem: true, drop all capabilities, disallow privileged: true), use private registries (Artifact Registry), pin image digests (not tags), rebuild regularly to include CVE fixes from base image updates.

Q258.

What are service account security best practices?

Answer:

Create one service account per workload/application (not shared). Grant minimum required IAM roles at lowest resource scope. Never create service account keys unless absolutely required – use Workload Identity (GKE), ADC metadata server (VMs/Cloud Run), or Workload Identity Federation (external workloads). If keys are required: store in Secret Manager, rotate every 90 days, delete unused keys, monitor usage in Cloud Audit Logs. Set iam.disableServiceAccountKeyCreation org policy. Audit with IAM Recommender quarterly.

Q259.

What is Artifact Registry and its security features?

Answer:

Artifact Registry is GCP's managed repository for containers and language packages. Security features: (1) IAM access control per repository (fine-grained, not project-wide like old Container Registry); (2) Vulnerability scanning via Container Analysis API — scans OS packages and language dependencies (Maven, npm, Python) for known CVEs; (3) VPC Service Controls integration (prevent exfiltration); (4) CMEK encryption; (5) Binary Authorization attestation for signed image verification; (6) Regional storage for data residency. Replaces Container Registry (gcr.io).

Q260.

What is GCP's approach to GDPR compliance?

Answer:

GCP supports GDPR through: (1) Data Processing Agreement (DPA) — sign Google's DPA via Cloud Console; (2) Data residency — restrict resources to EU regions via Org Policy; (3) Data subject rights — use DLP API to find and delete personal data, BigQuery table expiration; (4) Breach notification — 72-hour notification requirement; (5) Processor accountability — Google's ISO 27001, SOC 2 certifications; (6) Encryption — CMEK for customer data control; (7) Audit logging — Cloud Audit Logs track all data access. Review GCP's GDPR Compliance Documentation regularly.

Q261.

What is Cloud IDS?

Answer:

Cloud IDS (Intrusion Detection System) is a managed network threat detection service powered by Palo Alto Networks threat intelligence. It uses Packet Mirroring to inspect network traffic for: malware C2 communications, network exploits, port scanning, spyware, and lateral movement indicators. Alerts appear in Cloud Logging with threat severity, source/destination, and recommended actions. Integrates with SCC. Deployment: create IDS endpoint in a zone, attach Packet Mirroring policy to inspect traffic from target VMs or subnets.

Q262.

What is Policy Intelligence in GCP?

Answer:

Policy Intelligence tools: (1) IAM Recommender — ML-based suggestions to remove excess IAM permissions; (2) Policy Troubleshooter — diagnose why a principal can or cannot perform an action (simulates IAM evaluation); (3) Policy Analyzer — answer 'who has access to resource X?' or 'what resources can principal Y access?' at scale across the org; (4) Firewall Insights — identify unused, shadowed, or overly permissive firewall rules with usage metrics; (5) Role Recommender — suggest predefined role replacements for custom roles.

Q263.

What is Cloud Armor Adaptive Protection?

Answer:

Cloud Armor Adaptive Protection uses machine learning to establish normal traffic baselines for each protected backend service and detect volumetric L7 DDoS attacks. When an attack is detected: Adaptive Protection generates a suggested WAF rule to block the attack traffic (based on attack signatures — specific user agents, request patterns) and optionally auto-deploys the rule. Provides real-time attack telemetry and analytics in Cloud Logging. Available with Cloud Armor Managed Protection Plus tier subscription.

Q264.

What are VPC Service Controls security perimeters?

Answer:

VPC Service Controls perimeters enforce that API calls to protected GCP services (BigQuery, Cloud Storage, Spanner) originate only from within the defined perimeter or meet access level conditions. Even users with valid IAM permissions outside the perimeter are blocked (Context-Dependent Access denied). Protects against: data exfiltration via compromised credentials, unauthorized access from outside the organization's network, accidental public sharing, and insider threats using credentials from unauthorized locations.

Q265.

What is Chronicle SIEM?

Answer:

Chronicle is Google's cloud-native Security Information and Event Management (SIEM) built on Google's infrastructure for petabyte-scale security telemetry ingestion. Key capabilities: ingest logs from GCP (Cloud Audit Logs, VPC Flow Logs, SCC findings), on-premises, and third-party sources. Detection: YARA-L rules for behavioral threat detection. Investigation: Unified Data Model (UDM) normalizes all log sources for consistent querying. Retroactive search: re-run new detection rules against 12 months of historical data instantly. Integrates with SOAR platforms for automated response.

Q266.

What is the purpose of Data Access audit logs?

Answer:

Data Access logs record API calls that read resource configuration/metadata or create/modify/read user-provided resource data. Types: ADMIN_READ (read resource configuration), DATA_READ (read user data — e.g., reading a BigQuery table), DATA_WRITE (write user data). Disabled by default because they generate enormous log volume and cost. Enable selectively per service via gcloud logging write or Console □ Audit Logs for services handling sensitive data (BigQuery, Cloud SQL, Cloud Storage) to satisfy compliance requirements.

Q267.

What are HIPAA-eligible services on GCP?

Answer:

Google supports HIPAA compliance for covered entities and business associates via a signed Business Associate Agreement (BAA). HIPAA-eligible services include: Compute Engine, GKE, App Engine, Cloud Run, Cloud Functions, Cloud Storage, Cloud SQL, Cloud Spanner, BigQuery, Cloud KMS, Cloud Logging, Cloud Monitoring, Pub/Sub, Healthcare API, Apigee, and others. Not all GCP services are HIPAA-eligible — check Google's HIPAA compliance documentation for the current list before processing PHI with any service.

Q268.

What is Supply Chain Security in GCP?

Answer:

Supply chain security protects the software delivery pipeline from code to production. GCP tools: (1) Artifact Registry vulnerability scanning — scan container images and packages for CVEs. (2) Binary Authorization — cryptographic attestation gates: only images attested by authorized build systems deploy. (3) Cloud Build provenance — automatically generates SLSA provenance attestations for builds (who built it, from what source, with what tools). (4) Software Delivery Shield — end-to-end supply chain security dashboard showing SLSA level, vulnerability posture, and policy compliance across Cloud Build, Artifact Registry, and GKE.

Q269.

What is GCP's approach to Zero Trust security?

Answer:

GCP's Zero Trust implementation: never trust, always verify — every access request authenticated and authorized regardless of network origin. Components: (1) Cloud IAP — identity and context-based application access without VPN. (2) BeyondCorp Enterprise — device trust via Endpoint Verification, context-aware access policies. (3) Access Context Manager — access levels based on device health, IP, identity, and location. (4) VPC Service Controls — data-plane perimeter protection. (5) Cloud Armor — edge security with WAF. (6) mTLS via ASM — mutual authentication between every microservice. Eliminates the assumption that internal network = trusted.

Q270.

What is the Google Cloud Security Foundations Blueprint?

Answer:

The Security Foundations Blueprint is Google's opinionated, production-ready reference architecture for a secure GCP organization. It implements: multi-project hierarchy with Shared VPC, centralized logging to a security logging project, org-level IAM policies (restrict basic roles, SA key creation, public access), VPC Service Controls perimeters, centralized KMS, Cloud Armor policies, and SCC Premium configuration. Deployable via Terraform. Use as starting point for enterprise GCP deployments instead of manually implementing each security control. Available at cloud.google.com/architecture/security-foundations.

Section 10: Real-world Scenario-based Questions

30 questions with comprehensive, interview-focused answers.

Q271.

Your team needs to migrate a monolithic on-premises application to GCP. What is your migration strategy?

Answer:

Use the 6 Rs framework, choosing per application component: (1) Rehost (lift-and-shift): migrate VMs directly to Compute Engine with Migrate to Virtual Machines — fastest, minimal code change, limited cloud benefit. (2) Replatform: move to managed services (Cloud SQL instead of self-managed MySQL, App Engine instead of managing Apache) — moderate effort, operational savings. (3) Refactor/Re-architect: redesign as microservices on GKE or Cloud Run — maximum cloud benefit, highest effort. Strategy: assess each component (dependencies, latency, data volume), plan migration waves (start with least critical), use Database Migration Service for databases, run parallel environments during cutover, and validate with load testing before traffic switch.

Q272.

A production GKE cluster is experiencing high latency. How do you troubleshoot?

Answer:

Systematic approach: (1) Cloud Monitoring GKE Dashboard — check cluster CPU, memory, pod count. (2) `kubectl top nodes / kubectl top pods` — identify resource-saturated nodes or pods. (3) HPA status: `kubectl get hpa` — is the application scaling? Any pending pods? `kubectl describe pod` for pending pods to find scheduling failures. (4) Cloud Logging: filter errors at container level for application exceptions. (5) Cloud Trace: identify which service layer (app, cache, DB) is slow. (6) Database: Cloud SQL Insights for slow queries; Memorystore latency metrics. (7) Network: VPC Flow Logs for packet loss, Cloud NAT metrics for port exhaustion. (8) Node health: `kubectl get nodes` — any NotReady? Check node conditions with `kubectl describe node`.

Q273.

A Cloud Storage bucket with sensitive data was accidentally made public. What do you do?

Answer:

Immediate response: (1) Remove public access: `gcloud storage buckets remove-iam-policy-binding gs://BUCKET --member=allUsers --role=roles/storage.objectViewer`. Enable uniform bucket-level access if not already on. (2) Investigate: Cloud Audit Logs — who made it public and when? `gsutil ls -r` to inventory exposed objects. Cloud Storage access logs — what was downloaded? (3) DLP API scan — classify what types of sensitive data were exposed. (4) If PII/PHI exposed: initiate breach notification (GDPR: 72 hours). (5) Post-incident: implement constraints/storage.publicAccessPrevention org policy, SCC alert for public bucket findings, review all bucket permissions across org with Cloud Asset Inventory.

Q274.

Your application on Compute Engine needs to access Cloud SQL and Cloud Storage securely. Design the architecture.

Answer:

Secure design: (1) VM: create dedicated service account (`sa-myapp@PROJECT.iam.gserviceaccount.com`), grant `roles/cloudsql.client` and `roles/storage.objectAdmin` on specific bucket only. Attach SA to VM — no key files needed. (2) Network: enable Private Google Access on subnet, deploy VM without external IP, use Cloud NAT for outbound internet. Cloud SQL via private IP in the same VPC. (3) Cloud SQL: use Cloud SQL Auth Proxy sidecar or connect to private IP directly. App connects to `127.0.0.1:5432` (proxy handles auth + encryption). (4) Application code: use ADC — no credential configuration needed. DB password from Secret Manager (SA has `roles/secretmanager.secretAccessor`). (5) Firewall: allow subnet CIDR to Cloud SQL port. Deny all else.

Q275.

How would you design a highly available web application on GCP serving global users?

Answer:

Multi-region HA architecture: (1) DNS: Cloud DNS with low TTL (60s), failover health-checked records or let LB handle globally. (2) Load Balancing: Global HTTP(S) LB with Cloud Armor WAF, anycast IP routes users to nearest backend. (3) Compute: Regional MIGs in us-central1 + europe-west1 + asia-east1 with auto-scaling (min 2 instances per zone). (4) Cache: Cloud CDN for static assets (images, JS, CSS) at Google Edge PoPs. Memorystore (Redis) for session and application cache per region. (5) Database: Cloud Spanner multi-region (nam6 or eur3 config) for globally consistent ACID transactions. Cloud Storage multi-region for assets. (6) Monitoring: Cloud Monitoring uptime checks from all regions, alert on P99 latency and error rate. (7) DR: quarterly DR drill, RTO < 15 min, RPO < 1 min with Spanner.

Q276.

A Cloud Function is failing intermittently. How do you investigate?

Answer:

Structured investigation: (1) Cloud Logging: `resource.type='cloud_function'` `resource.labels.function_name='FUNC'` `severity>=ERROR` — find error messages and stack traces. Note timing patterns (time of day, frequency). (2) Error Reporting: aggregated error count trend, sample payloads. (3) Cloud Monitoring: function metrics — `error_count`, `execution_count`, `execution_times` (p99 latency spike?), `active_instances` (hitting limit?). (4) Memory: is `memory_usage` approaching limit? (5) Timeout: are invocations timing out? Downstream API slow? (6) Cold starts: large invocation latency spikes? Add `--min-instances=1`. (7) Dependencies: Cloud SQL connection limits? External API rate limiting? Check VPC connector if using private resources. (8) Retry storm: Pub/Sub retrying failed messages creating cascading load? Check for duplicate processing.

Q277.

How would you design a CI/CD pipeline for deploying to GKE?

Answer:

Pipeline with Cloud Build: (1) Source trigger: Cloud Build trigger on PR (plan) and merge to main (deploy). Integrate GitHub/GitLab. (2) Build stage: docker build --tag gcr.io/PROJECT/APP:\$COMMIT_SHA, run unit tests in parallel. (3) Security gates: Artifact Registry vulnerability scan — fail build if CRITICAL CVEs. Binary Authorization attestation after clean scan. (4) Deploy staging: kubectl apply -f k8s/ --namespace=staging or helm upgrade --install in staging cluster. Run integration/smoke tests. (5) Approval: Cloud Build manual approval step or Deployment Manager gate. (6) Deploy production: kubectl rollout update (maxSurge=2, maxUnavailable=0). Monitor error rate in Cloud Monitoring during rollout. (7) Automated rollback: Monitoring alert → Pub/Sub → Cloud Function → kubectl rollout undo if error rate spikes. (8) Notifications: Slack on success/failure via Cloud Build webhook.

Q278.

How would you implement cost optimization for a GCP environment with 200+ VMs?

Answer:

Structured cost optimization: (1) Analysis: billing export to BigQuery → Looker Studio dashboard by project/label/service. Identify top 20% resources consuming 80% of cost. (2) VM rightsizing: Compute Engine Recommender → identify over-provisioned VMs → resize during maintenance window. (3) Committed Use Discounts: analyze 3-month baseline usage → purchase 1-year CUDs for stable workloads → up to 37% savings. (4) Spot VMs: migrate dev/test, batch, CI/CD agents to Spot (up to 91% savings). (5) Instance scheduling: Resource Policies to stop dev VMs 7pm-8am weekdays and weekends → 70% savings for non-prod. (6) Storage: lifecycle rules on Cloud Storage buckets, delete unattached disks and unused snapshots (Asset Inventory). (7) Network: review Cloud NAT port allocation, optimize data transfer with CDN. (8) Labels: enforce labels via Org Policy, attribute costs to teams for accountability.

Q279.

A developer accidentally deleted a Cloud SQL database. How do you recover?

Answer:

Recovery playbook: (1) Assess: Is PITR enabled? When did the deletion occur? (2) PITR recovery (if enabled): `gcloud sql instances clone SOURCE_INSTANCE RECOVERY_INSTANCE --point-in-time='2024-01-15T14:25:00.000Z'` (timestamp just before deletion). Verify data completeness. (3) Backup restore (if no PITR): `gcloud sql backups restore BACKUP_ID --restore-instance=INSTANCE_ID`. Restores to last automated backup (may lose hours of data). (4) Verify restored data and application connectivity. Promote clone or update connection strings. (5) Root cause: Cloud Audit Logs — who ran the deletion, what client/IP? (6) Immediate prevention: enable PITR on all production instances, set 7-day binary log retention, restrict roles/cloudsql.admin to DBA team only, consider Cloud SQL Deletion Protection flag. (7) Document incident, create runbook for future recovery.

Q280.

How do you implement zero-downtime deployment on GKE?

Answer:

Zero-downtime configuration: (1) Deployment strategy: `spec.strategy.type: RollingUpdate` with `maxSurge: 1` (add one new pod before removing old), `maxUnavailable: 0` (never reduce below desired count). (2) Readiness Probe: configure HTTP readiness check — LB only routes to pods passing the probe. New pod not served traffic until it's truly ready. (3) Pod Disruption Budget: `minAvailable: 70%` ensures adequate capacity during rollout. (4) Graceful termination: `containerPort` receives `SIGTERM` on eviction. Set `terminationGracePeriodSeconds: 60`. Implement signal handler to stop accepting new requests and drain in-flight. (5) PreStop hook: sleep 10 seconds before `SIGTERM` to allow LB to remove pod from rotation. (6) Monitor: watch `error_rate_5xx` in Cloud Monitoring during rollout; trigger auto-rollback if error rate > 1%. (7) Canary option: deploy to 10% via traffic splitting, validate, then proceed.

Q281.

Your application needs to process 1 million IoT sensor readings per day. Design the architecture.

Answer:

IoT pipeline on GCP: (1) Ingestion: IoT devices → Pub/Sub topic (scalable, durable, handles any ingestion rate). Cloud IoT Core (if device management/MQTT needed). 1M/day = ~12/second — trivial for Pub/Sub. (2) Stream processing: Dataflow (Apache Beam) reads from Pub/Sub subscription — validates schemas, enriches with device metadata, aggregates 5-minute windows. (3) Hot path (real-time alerts): Dataflow → Pub/Sub alert topic → Cloud Functions → send alerts if sensor > threshold. (4) Warm path (operational dashboard): Dataflow → Bigtable (time-series key: deviceId#reverse_timestamp, read recent data in milliseconds for dashboards). (5) Cold path (analytics/ML): Dataflow → BigQuery (partitioned by date, clustered by device_type — SQL analytics, BQML for anomaly detection). (6) Visualization: Looker Studio → BigQuery (historical), custom React dashboard → Bigtable (real-time). (7) Monitoring: Pub/Sub oldest unacked message age alert (pipeline lag indicator).

Q282.

How would you set up a multi-project GCP organization with proper governance?

Answer:

Organization setup: (1) Hierarchy: Org Node → Folders (Production, Non-Production, Shared-Services) → Projects. Shared-Services: networking, logging, security, CI/CD. (2) Identity: Cloud Identity, separate break-glass admin accounts from daily-use accounts, enforce MFA via Cloud Identity settings. (3) Billing: separate billing accounts per business unit for chargeback. (4) Network: Shared VPC — host project in Shared-Services folder owned by network team. Service projects attach to host VPC subnets. Private Google Access and Cloud NAT in host VPC. (5) IAM: grant team roles at folder level (developers: Editor on Non-Prod folder, Viewer on Production). (6) Org Policies: restrict external IPs, enforce uniform bucket access, restrict SA key creation, limit to approved regions. (7) Logging: org-level sink to centralized logging project (BigQuery + GCS). (8) Security: SCC at org level, org-wide Cloud Armor policies, centralized KMS keyring. (9) Automation: Terraform for project factory, Cloud Build for deployments.

Q283.

Your Cloud SQL instance is experiencing slow queries. How do you diagnose and fix?

Answer:

Diagnosis: (1) Query Insights (Console ▢ Cloud SQL ▢ Query Insights): top queries by normalized latency and execution count, execution plan visualization — identify missing indexes and full table scans. (2) Slow query log: enable via flags (`slow_query_log=on`, `long_query_time=1` for MySQL) — logs appear in Cloud Logging. (3) Cloud Monitoring: CPU saturation, disk IOPS hitting limit, connections near `max_connections`. (4) `EXPLAIN ANALYZE` on top slow queries — look for Seq Scan on large tables, high row estimates, nested loop joins. Fixes: (1) Add indexes for WHERE, JOIN, ORDER BY columns. (2) Rewrite N+1 queries (batch fetches). (3) Connection pooling via Cloud SQL Auth Proxy or PgBouncer (reduce connection overhead). (4) Read replicas for reporting/analytics queries. (5) Upgrade machine type if CPU-saturated. (6) Partitioning large tables by date for time-range queries.

Q284.

How would you implement disaster recovery for a critical application?

Answer:

DR strategy (RTO < 15 min, RPO < 1 min): (1) Multi-region architecture: active primary in us-central1, warm standby in us-east1. (2) Compute: regional MIGs in both regions attached to Global HTTP(S) LB. LB health check automatically fails over if primary region backends fail. (3) Database: Cloud Spanner with `nam6` multi-region config (automatic synchronous replication, 99.999% SLA, zero RPO on regional failure). Or Cloud SQL cross-region replica (async, ~seconds RPO) promoted on failure. (4) Storage: Cloud Storage dual-region (synchronous geo-redundancy). (5) DNS: Cloud DNS with health-checked failover (60s TTL). (6) Failover runbook: documented, tested, <10 steps. Automate with Cloud Functions triggered by Monitoring alerts. (7) Testing: quarterly DR drills — actually fail over, measure actual RTO/RPO, update runbooks. (8) On-call: PagerDuty integration, clear escalation path, DR contacts listed in runbook.

Q285.

How do you troubleshoot an unresponsive Compute Engine VM?

Answer:

Troubleshooting steps: (1) Check VM status in Console — RUNNING? Any recent operations (maintenance migration)? (2) Serial console output: `gcloud compute instances get-serial-port-output INSTANCE --zone=ZONE` — kernel panic? Filesystem error? OOM kill? Boot failure? (3) IAP SSH: `gcloud compute ssh INSTANCE --tunnel-through-iap` — bypasses need for external IP and SSH firewall rule. (4) Check firewall rules: is port 22 open from your IP? Does network tag/SA match? (5) Cloud Monitoring: CPU 100%? Memory exhausted? Disk full? (6) Disk full fix: stop VM, create snapshot, detach boot disk, attach as secondary to rescue VM, mount and clear space, reattach. (7) Kernel fix: stop VM, detach boot disk, attach to rescue VM, chroot, fix kernel/config, reattach. (8) Last resort: restore from latest snapshot to new VM instance.

Q286.

Design a data pipeline to load on-premises database data to BigQuery daily.

Answer:

ELT pipeline design: (1) Extraction: Datastream (CDC replication, sub-minute latency) streams Oracle/MySQL/PostgreSQL changes to Cloud Storage or directly to BigQuery. Or: extract script (JDBC → CSV) on a VM or on-premises server, upload to GCS via `gsutil`. (2) Storage: Cloud Storage staging bucket (parquet or CSV, partitioned by date). (3) Load: BigQuery Data Transfer Service schedules daily load from GCS → BigQuery table (or Dataflow for complex transformations). (4) Transform: dbt (data build tool) runs SQL transformations in BigQuery — clean, deduplicate, join dimension tables. (5) Orchestration: Cloud Composer (managed Airflow) DAG: extract → upload → load → transform → validate → notify. (6) Data quality: dbt tests (not null, unique, referential integrity) + Great Expectations. Alert on failures. (7) Monitoring: Composer DAG run status, BigQuery job completion, row count validation (compare source vs BigQuery count).

Q287.

How do you implement VPC peering between two projects?

Answer:

Setup process: (1) Project A: `gcloud compute networks peerings create peering-a-b --network=vpc-a --peer-project=PROJECT_B_ID --peer-network=vpc-b`. (2) Project B: `gcloud compute networks peerings create peering-b-a --network=vpc-b --peer-project=PROJECT_A_ID --peer-network=vpc-a`. Both peerings must be active for connectivity. (3) Verify: `gcloud compute networks peerings list --network=vpc-a` — state should be ACTIVE. (4) Firewall rules: update both VPCs to allow traffic from peer CIDR ranges. (5) Limitations to communicate: IP ranges must not overlap, peering is non-transitive ($A \nleftrightarrow B, B \nleftrightarrow C \neq A \leftrightarrow C$), max 25 peerings per VPC, no peering alongside Shared VPC. (6) Alternative for transitive routing: HA VPN Hub-and-Spoke or Network Connectivity Center.

Q288.

How do you move from console-based GCP management to Infrastructure as Code?

Answer:

IaC adoption roadmap: (1) Tool selection: Terraform (recommended — multi-cloud, largest community, Terraform Cloud/Enterprise for teams). (2) Pilot: start with non-critical dev environment, write Terraform for one project's VPC, Cloud SQL, GKE. (3) Import existing resources: `terraform import google_compute_instance.vm projects/P/zones/Z/instances/NAME` to bring existing infra under management without recreating. (4) State backend: GCS bucket with versioning for Terraform state, enable state locking (GCS built-in). (5) GitOps workflow: Git repo for all Terraform, Cloud Build pipeline: `PR → terraform plan` (post to PR), `merge → terraform apply`. Require PR approval for production changes. (6) Modules: create reusable modules for standard patterns (gke-cluster, vpc-network, cloud-sql). (7) Policy: enforce 'no manual changes' rule — all changes via Terraform PRs. (8) Gradual migration: import resources project by project, train team on HCL syntax.

Q289.

How would you implement PCI DSS compliance for a payment application on GCP?

Answer:

PCI DSS on GCP — key controls: (1) Network segmentation: separate GCP project + VPC for cardholder data environment (CDE). VPC Service Controls perimeter around CDE services (Cloud Storage, BigQuery, Cloud SQL holding card data). (2) Firewall: no direct internet to CDE VMs. All inbound via WAF (Cloud Armor with OWASP rules). Restrict outbound to specific IPs/ports. (3) Data protection: tokenize PANs (replace card numbers with tokens) to minimize CDE scope. CMEK for all CDE storage. DLP API to detect PAN leakage in logs. (4) Access control: MFA enforced for all admin access. OS Login for SSH. Time-bound IAM Conditions for emergency access. (5) Monitoring: all audit logs enabled and exported (90-day+ retention). Cloud IDS for threat detection. SCC for misconfigurations. (6) Testing: quarterly ASV vulnerability scans (Web Security Scanner), annual penetration test, quarterly firewall rule review. (7) Governance: sign Google's PCI DSS BAA, reference Google's PCI DSS Attestation of Compliance (AoC).

Q290.

Design an ML model serving architecture on GCP for high-traffic predictions.

Answer:

Scalable ML serving on GCP: (1) Model training: Vertex AI Training with custom containers or pre-built TensorFlow/PyTorch environments. Use GPUs (A100) for training. (2) Model registry: Vertex AI Model Registry for versioning, metadata, and lineage tracking. (3) Online serving: Vertex AI Prediction endpoints (managed, auto-scales, canary between model versions, private endpoint option). For extreme performance: deploy to Cloud Run (containerized TF Serving/TorchServe) with pre-provisioned concurrency. (4) Caching: Memorystore (Redis) cache prediction results for repeated identical inputs — reduces model serving load and latency. (5) Feature consistency: Vertex AI Feature Store for feature serving in both training and prediction — prevents training-serving skew. (6) Monitoring: Vertex AI Model Monitoring detects input data drift and prediction drift. Alert and trigger automatic retraining via Vertex AI Pipelines. (7) Batch: Vertex AI Batch Prediction for offline scoring at scale.

Q291.

How do you manage secret rotation for Cloud Run services?

Answer:

Secret rotation strategy: (1) Store all secrets in Secret Manager — never environment variables or config files. (2) Reference in Cloud Run: `--set-secrets=DB_PASSWORD=my-db-secret:latest` (latest version auto-fetched on new revision deploy). (3) Rotation process: create new secret version (`gcloud secrets versions add SECRET_NAME --data-file=newpassword.txt`). New Cloud Run revisions automatically use the new version. (4) Zero-downtime rotation: application should support both old and new credential simultaneously during rotation window (DB accepts both passwords briefly). (5) Automated rotation: Secret Manager rotation schedule □ Pub/Sub notification □ Cloud Function □ update backend system password □ add new Secret Manager version □ disable old version. (6) Verify: test new secret works in staging before disabling old version. (7) Audit: Cloud Audit Logs records all secret access — detect unauthorized access immediately.

Q292.

How do you implement centralized logging across 50 GCP projects?

Answer:

Centralized logging architecture: (1) Org-level aggregate sink: `gcloud logging sinks create org-sink bigquery.googleapis.com/projects/LOGGING_PROJECT/datasets/org_logs --organization=ORG_ID --include-children --log-filter='severity>=WARNING'`. Captures all logs from all 50 projects automatically. (2) Multiple destinations: BigQuery (analytics, SQL queries, 90+ day retention), Cloud Storage (cold archival for compliance, 7+ year retention), Pub/Sub (real-time streaming to SIEM/Chronicle). (3) Cost management: exclude noisy/low-value logs per project (dev debug logs, health check noise) with exclusion filters. Estimate volume before enabling Data Access logs org-wide. (4) Access control: BigQuery authorized views allow teams to query only their own project's logs. (5) Centralized alerting: Monitoring workspace scoping all 50 projects for unified alerting. (6) Compliance: lock Cloud Storage archival bucket with retention policy for WORM compliance. Validate log completeness with log-based metric counting log entries per project.

Q293.

How do you implement a multi-tenant SaaS on GCP?

Answer:

Multi-tenancy options: (1) Shared infrastructure (pool model): all tenants share GKE cluster, Cloud SQL (separate database per tenant), Cloud Storage (separate bucket per tenant). Cost-efficient, use Kubernetes namespaces + network policies for isolation. Implement tenant ID in all requests; validate server-side. (2) Per-tenant infrastructure (silo model): separate GCP project per tenant for strongest isolation. Automate with Terraform project factory + Cloud Build. Higher cost but simplifies compliance (HIPAA, PCI per-tenant isolation). (3) Hybrid: shared compute (GKE multi-tenant), isolated data (separate Cloud SQL per tenant with CMEK per tenant). (4) Key concerns: tenant data isolation (row-level security in DB, separate KMS keys), billing attribution (labels per tenant in billing export), rate limiting per tenant (Cloud Armor per-IP, Apigee quotas per API key), tenant provisioning automation (Terraform + CI/CD pipeline for new tenant onboarding in < 30 minutes).

Q294.

How do you implement blue-green deployment on Cloud Run?

Answer:

Cloud Run blue-green: (1) Deploy green version with zero traffic: `gcloud run deploy SERVICE --image=NEW_IMAGE --no-traffic --tag=green`. New revision created but receives no requests. (2) Test green via tag URL: `https://green---SERVICE-HASH.run.app` — verify functionality without affecting production. (3) Gradual traffic shift: `gcloud run services update-traffic SERVICE --to-revisions=GREEN_REVISION=10,BLUE_REVISION=90` — monitor error rate for 15 minutes. (4) Full cutover: `gcloud run services update-traffic SERVICE --to-revisions=GREEN_REVISION=100`. (5) Instant rollback: `gcloud run services update-traffic SERVICE --to-revisions=BLUE_REVISION=100` — takes effect in seconds. (6) Clean up: delete blue revision after validation period. Cloud Run keeps all revisions available for instant rollback. No infrastructure changes needed — purely traffic weight management.

Q295.

How do you implement rate limiting for a public API on GCP?

Answer:

Rate limiting stack: (1) Edge protection (Cloud Armor): IP-based rate limiting rule — ban IP after 1000 requests/minute. Protects against brute force and DDoS at Google's edge before reaching your infrastructure. (2) API Gateway (Apigee or Cloud Endpoints): per-API-key quotas (e.g., 100 req/min for free tier, 10K req/min for premium), burst limits, usage plans. Returns 429 Too Many Requests with Retry-After header. (3) Application-level (Memorystore Redis): token bucket or sliding window counter per user ID for fine-grained limiting when multiple API keys belong to one account. (4) GKE Ingress + Envoy: service mesh-level rate limiting via Envoy rate limit filter for east-west (service-to-service) rate limiting. (5) Monitoring: Cloud Monitoring for 429 response rate, Cloud Armor denied request metrics. Alert when rate limiting is blocking legitimate traffic. Adjust limits based on data.

Q296.

Your startup is running out of Cloud SQL connections under peak load. How do you fix this?

Answer:

Connection exhaustion fix: (1) Diagnose: Cloud SQL Monitoring — database/postgresql/num_backends (PostgreSQL) or database/mysql/threads_connected (MySQL) approaching max_connections. (2) Connection pooling: deploy PgBouncer (PostgreSQL) in transaction mode as a Deployment in GKE or on a VM. PgBouncer multiplexes 1000 app connections into 50 DB connections. Or use Cloud SQL Auth Proxy built-in connection pooling (--max-connections flag). (3) Cloud Run/Functions: enable Cloud SQL Auth Proxy connection pooling — serverless functions spawn many connections without pooling. (4) Increase max_connections: Cloud SQL flag max_connections (requires instance restart). Costs more memory per connection. (5) Read replicas: route read-only queries to replicas — reduces primary load. (6) Connection validation: implement connection health checks in app code; remove stale connections. (7) Monitor: alert when connections > 80% of max_connections.

Q297.

How would you migrate a stateful application with a database from on-premises to GCP with minimal downtime?

Answer:

Migration strategy for minimal downtime: (1) Assessment: map all dependencies (DB version, stored procedures, triggers, linked servers), measure data size and replication bandwidth. (2) Parallel run setup: provision Cloud SQL (same DB version) in GCP. Use Database Migration Service (DMS) for continuous replication: set up CDC replication, let initial load complete (days for large DBs), verify replica lag < 30 seconds. (3) Application migration: deploy app to Compute Engine or GKE (behind Cloud Load Balancer) pointing to on-premises DB still. (4) Test window: run GCP app in shadow mode, compare query results against on-premises. (5) Cutover (< 5 minutes): (a) Put on-premises app in maintenance mode. (b) Wait for DMS replica to catch up (lag = 0). (c) Promote Cloud SQL replica to primary. (d) Update GCP app DB connection string to Cloud SQL. (e) Switch DNS/load balancer to GCP. (f) Monitor errors. (6) Rollback plan: Cloud SQL export to on-premises if critical issues. (7) Decommission: after 2 weeks stable, decommission on-premises DB.

Q298.

How do you implement event-driven microservices architecture on GCP?

Answer:

Event-driven microservices with GCP: (1) Event bus: Pub/Sub as the central event broker — durable, scalable, supports push/pull, dead-letter topics, ordering within message keys. (2) Service communication: services publish events (OrderPlaced, PaymentProcessed) to Pub/Sub topics; interested services subscribe. Services are decoupled — producer doesn't know consumers. (3) Compute: Cloud Run (scale to zero, fast scaling for event spikes) or GKE (complex services needing persistent connections). (4) Choreography pattern: each service reacts to events independently. Orchestration pattern: Cloud Workflows coordinates multi-step processes with state. (5) Error handling: dead-letter topics for failed messages after N retries. Cloud Functions processes DLT and alerts/retries manually. (6) Event schema: use Pub/Sub schema registry (Avro/Protobuf) for schema evolution governance. (7) Observability: Distributed traces (Cloud Trace + OpenTelemetry) across service boundaries via trace context propagation in Pub/Sub message attributes.

Q299.

How do you set up a GCP environment for a new development team following security best practices?

Answer:

New team GCP onboarding: (1) Project structure: separate projects for dev, staging, production (different billing, IAM isolation). (2) IAM: create Google Group for the team (team-dev@company.com), grant roles/editor on dev project and roles/viewer on staging. No individual IAM bindings. Production: roles/viewer only, deploy only via CI/CD service account. (3) Network: custom-mode VPC (never default), Private Google Access enabled, no external IPs on VMs, Cloud NAT for outbound. (4) Org Policies: enforce on all projects – no public Cloud Storage, no SA key creation (use Workload Identity), restrict to approved regions. (5) CI/CD: Cloud Build service account with roles/container.developer and roles/run.admin for staging/prod deployments – developers push code, pipeline deploys. (6) Secrets: Secret Manager for all credentials, access via SA + Workload Identity. (7) Monitoring: SCC alerts for misconfigurations, Cloud Monitoring dashboards, budget alert at 80% of monthly budget. (8) Training: GCP fundamentals, security practices, cost awareness workshop before team starts.

Q300.

How do you handle a GCP outage affecting your production application?

Answer:

Outage response: (1) Detect: Cloud Monitoring alert fires (error rate spike, latency increase, uptime check failures). Verify on Google Cloud Status page (status.cloud.google.com) – is this a GCP regional outage? (2) Triage: identify affected components (single zone? single region? specific service?). Check SCC for any security events. (3) Failover (if multi-region): Global LB automatically redirects to healthy region if backends fail health checks. Verify failover is active in LB dashboard. Manual failover if needed: update Cloud DNS or LB backend weights. (4) Communication: post internal status update immediately (Slack, PagerDuty incident), update external status page if customer-facing. (5) Workarounds: disable non-critical features (circuit breakers), serve cached/static content from Cloud CDN, enable maintenance mode for affected services. (6) Monitor GCP status updates: subscribe to Cloud Status RSS feed. (7) Post-incident: document timeline, customer impact, failover effectiveness. File GCP support ticket for incident report and SLA credit. Review DR runbook gaps.